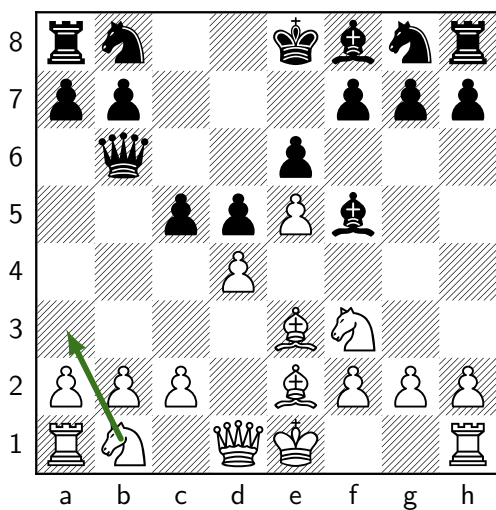


# Melanie-ANALOGY — Caro-Kann, Advance

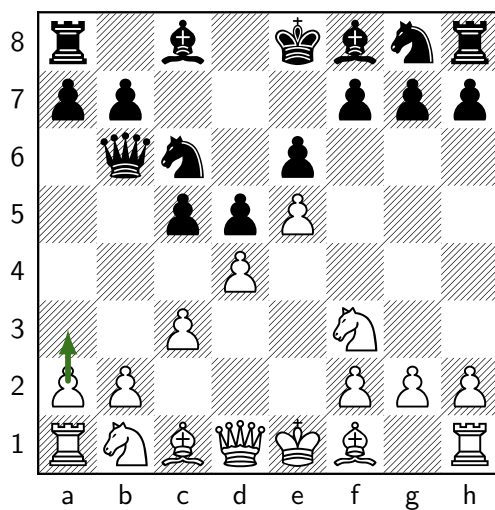
Melanie (Deep\_Vision)

## Abstract

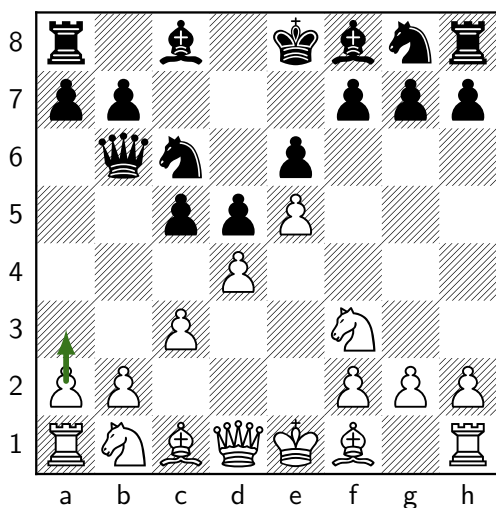
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



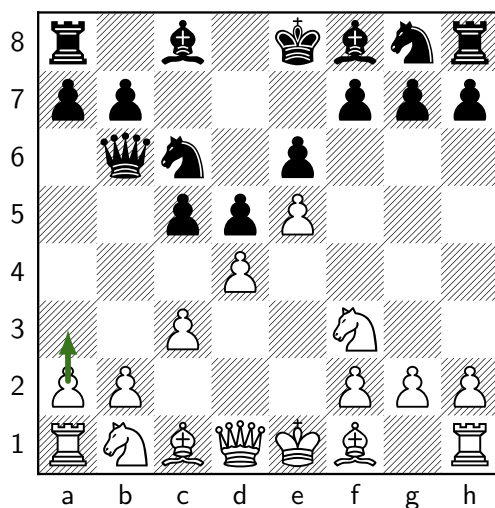
(a)



(b)

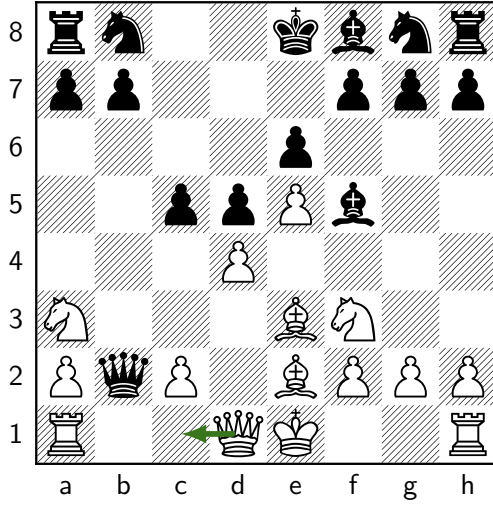


(c)

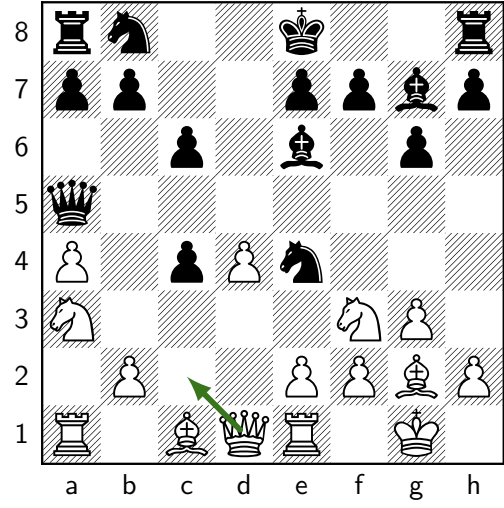


(d)

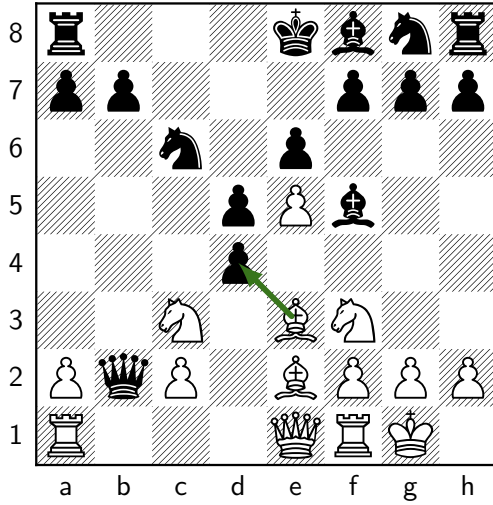
Figure 1: **Move 7, White: Na3.** (a) Our position; Melanie plays Na3 (b1a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2482` (ply 10) [structural distance  $d=55$ , QAP obj=93]: the master played a3; under  $\mu$  our Nb1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Pe5 as their Nb1, our Pa2 as their Bf1; transferred move b1a3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2483` (ply 10) [structural distance  $d=55$ , QAP obj=93]: the master played a3; under  $\mu$  our Nb1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Pe5 as their Nb1, our Pa2 as their Bf1; transferred move b1a3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2550` (ply 10) [structural distance  $d=55$ , QAP obj=93]: the master played a3; under  $\mu$  our Nb1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Pe5 as their Nb1, our Pa2 as their Bf1; transferred move b1a3 (**endorses this move**). Synthesis: 30 retrieved analogs transferred to Na3 (summed QAP objective 2792, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



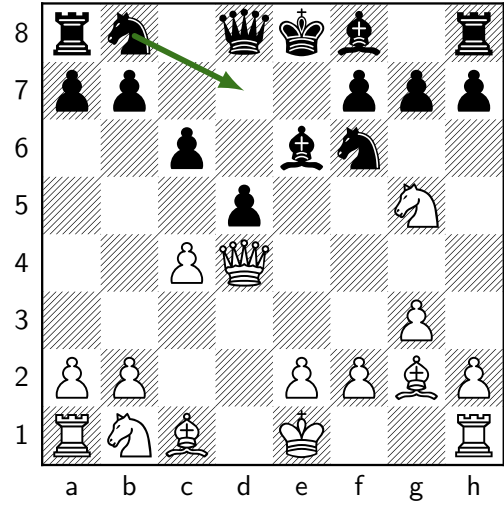
(a)



(b)

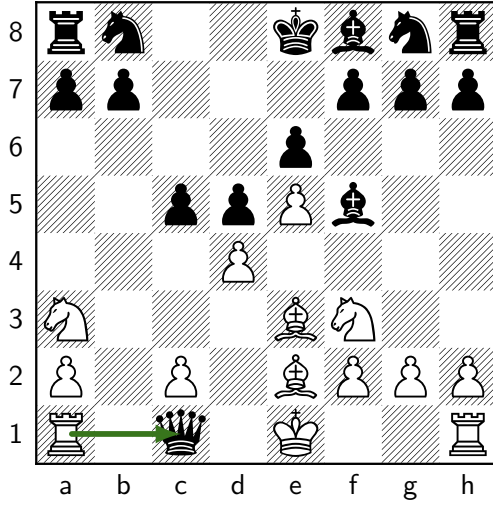


(c)

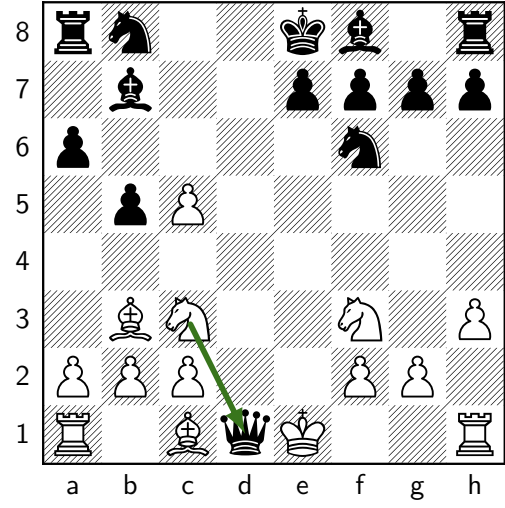


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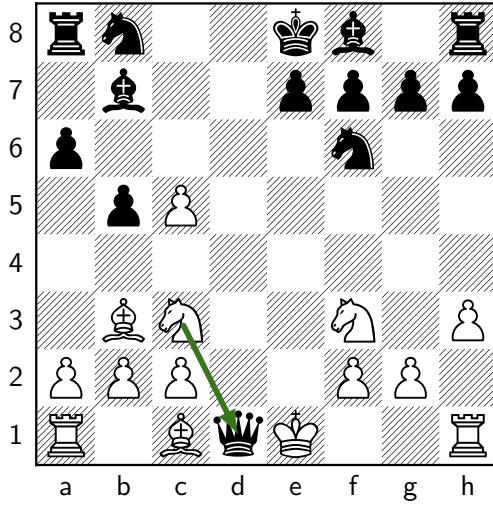
Figure 2: **Move 8, White: Qc1**. (a) Our position; Melanie plays Qc1 (d1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1524` (ply 18) [structural distance  $d=104$ , QAP obj=39]: the master played Qc2; under  $\mu$  our Qd1 plays the role of that game’s Qd1; cross-type roles: our ng8 as their Ra1, our Na3 as their Bg2; transferred move d1c1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2452` (ply 18) [structural distance  $d=87$ , QAP obj=137]: the master played Bxd4; under  $\mu$  our Ra1 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our qb2 as their pd4, our Be3 as their Pe5; transferred move a1c1. (d) Retrieved from `ccrl_top_p0002#617` (ply 13) [structural distance  $d=96$ , QAP obj=66]: the master played Nbd7; under  $\mu$  our Be2 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Be3 as their Ra8, our Nf3 as their Bf8; transferred move e2f1. Synthesis: 1 retrieved analog transferred to Qc1 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



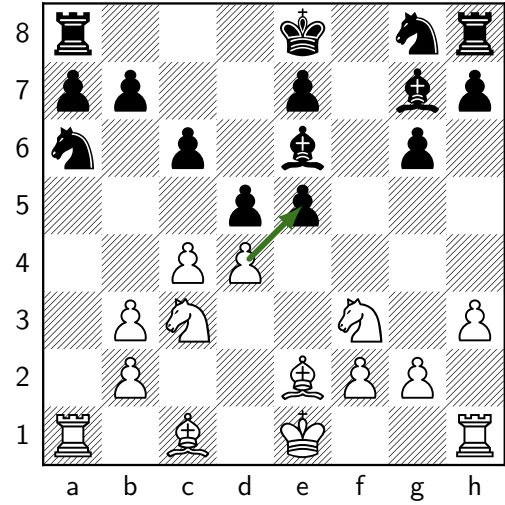
(a)



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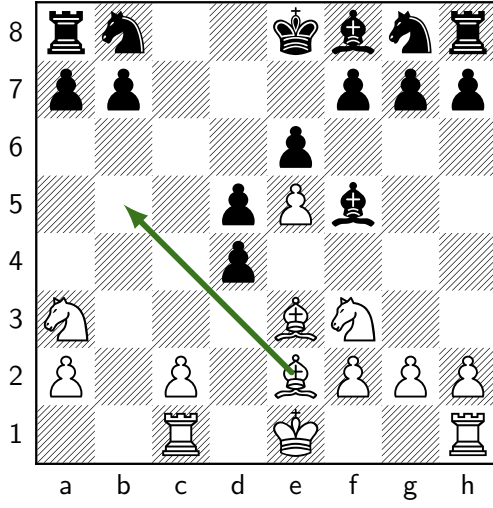


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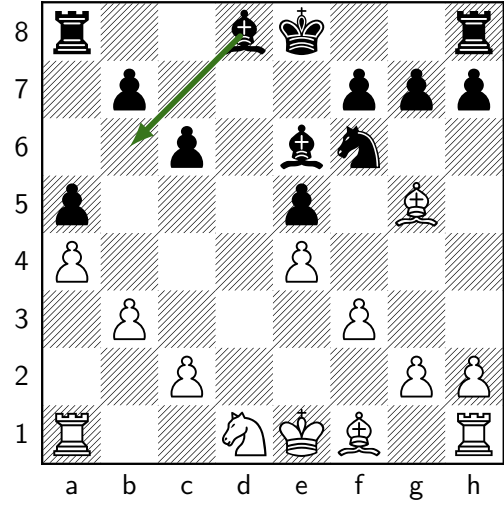


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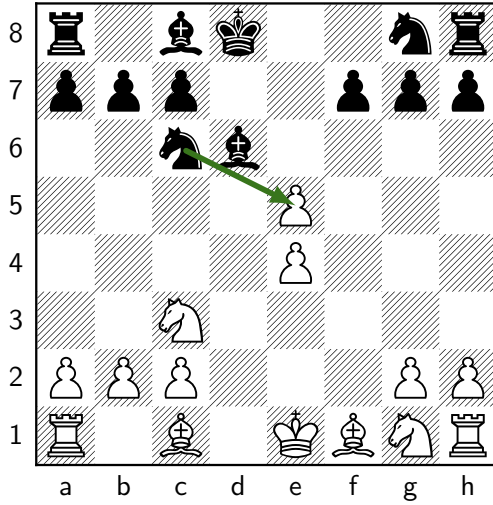
Figure 3: **Move 9, White: Rxc1.** (a) Our position; Melanie plays Rxc1 (a1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1494` (ply 18) [structural distance  $d=116$ , QAP obj=50]: the master played Nxd1; under  $\mu$  our *Ra1* plays the role of that game’s *Nc3* (a **cross-type** correspondence); cross-type roles: our *Pg2* as their *Ra1*, our *pd5* as their *Bc1*; transferred move a1c1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1495` (ply 18) [structural distance  $d=116$ , QAP obj=50]: the master played Nxd1; under  $\mu$  our *Ra1* plays the role of that game’s *Nc3* (a **cross-type** correspondence); cross-type roles: our *Pg2* as their *Ra1*, our *pd5* as their *Bc1*; transferred move a1c1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3483` (ply 20) [structural distance  $d=136$ , QAP obj=7]: the master played dxe5; under  $\mu$  our *Ra1* plays the role of that game’s *Pd4* (a **cross-type** correspondence); cross-type roles: our *Be2* as their *Ra1*, our *Na3* as their *Bc1*; transferred move a1c1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rxc1 (summed QAP objective 107, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



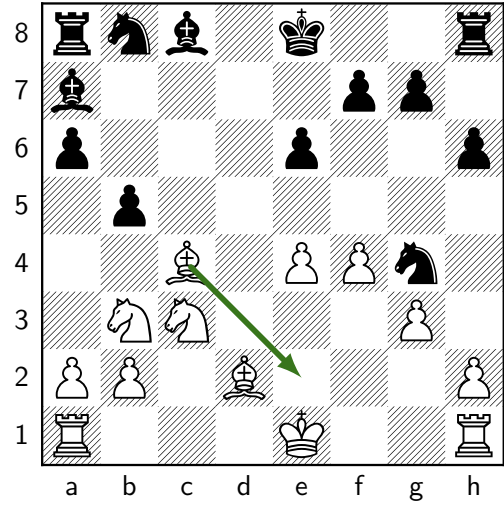
(a)



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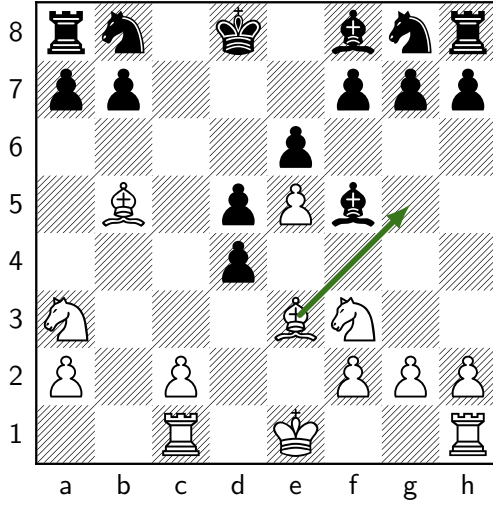


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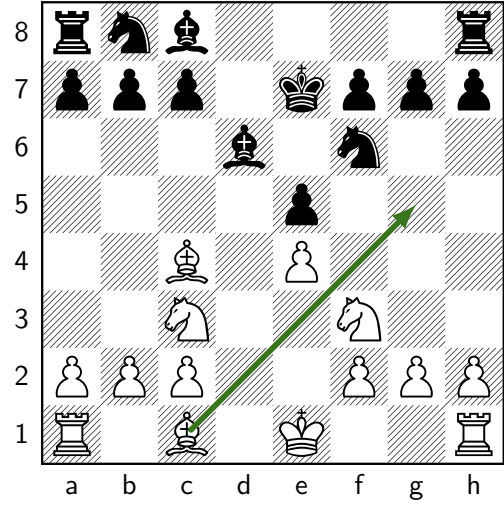


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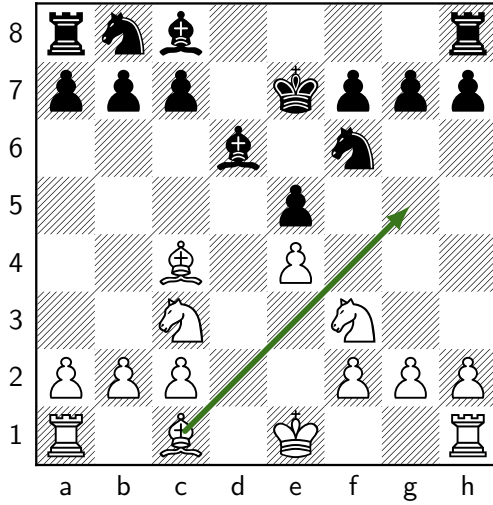
Figure 4: **Move 10, White: Bb5+.** (a) Our position; Melanie plays Bb5+ (e2b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3762` (ply 23) [structural distance  $d=71$ , QAP obj=25]: the master played Bb6; under  $\mu$  our *Be2* plays the role of that game’s *Bd1*; cross-type roles: our *pg7* as their *Ra8*, our *Nf3* as their *Be6*; transferred move e2b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2631` (ply 13) [structural distance  $d=70$ , QAP obj=76]: the master played Nxe5; under  $\mu$  our *Nf3* plays the role of that game’s *Nc3*; cross-type roles: our *Pe5* as their *Ra8*, our *pf7* as their *Rh8*; transferred move f3d4. (d) Retrieved from `ccr1_top_p0001#1574` (ply 26) [structural distance  $d=73$ , QAP obj=63]: the master played Be2; under  $\mu$  our *Be3* plays the role of that game’s *Bc4*; cross-type roles: our *Na3* as their *Rh1*, our *Pe5* as their *Nb3*; transferred move e3d2. Synthesis: 1 retrieved analog transferred to Bb5+ (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



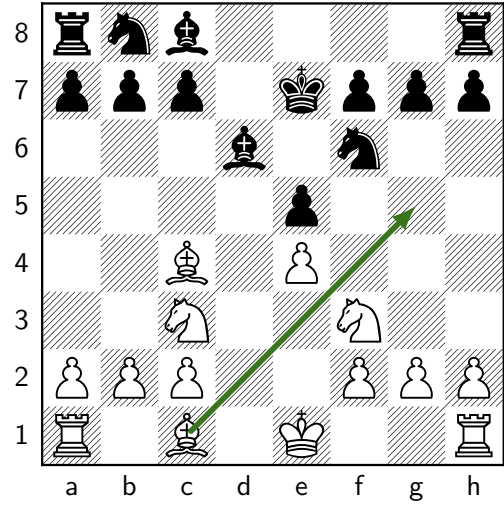
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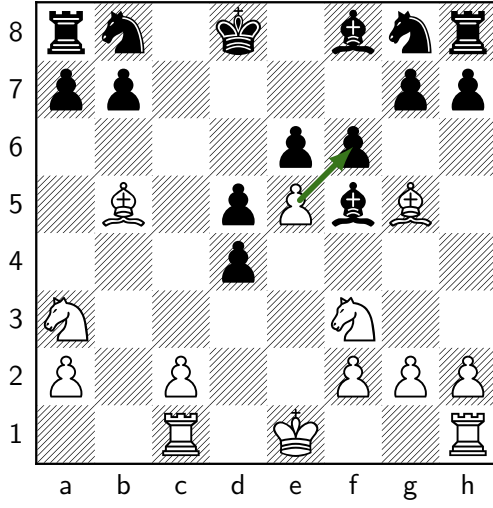


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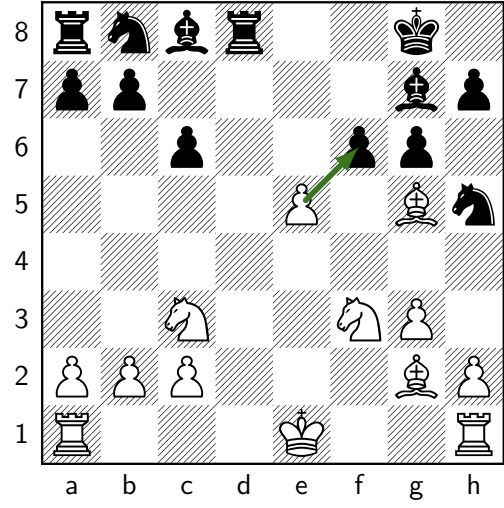


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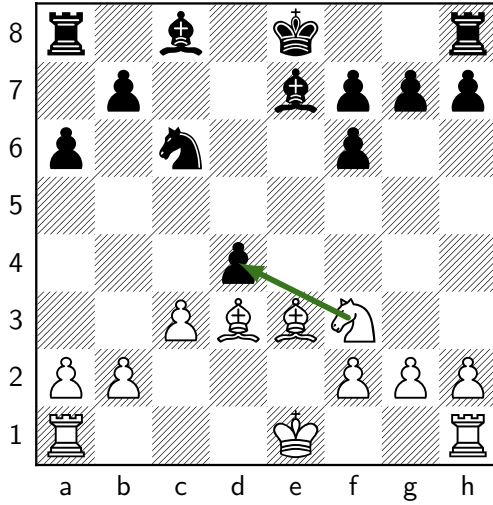
Figure 5: **Move 11, White: Bg5+.** (a) Our position; Melanie plays Bg5+ (e3g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1440` (ply 14) [structural distance  $d=78$ , QAP obj=3]: the master played Bg5; under  $\mu$  our *Be3* plays the role of that game’s *Bc1*; cross-type roles: our *nb8* as their *Pa2*, our *pd4* as their *Nc3*; transferred move e3g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1441` (ply 14) [structural distance  $d=78$ , QAP obj=3]: the master played Bg5; under  $\mu$  our *Be3* plays the role of that game’s *Bc1*; cross-type roles: our *nb8* as their *Pa2*, our *pd4* as their *Nc3*; transferred move e3g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2704` (ply 14) [structural distance  $d=78$ , QAP obj=3]: the master played Bg5; under  $\mu$  our *Be3* plays the role of that game’s *Bc1*; cross-type roles: our *nb8* as their *Pa2*, our *pd4* as their *Nc3*; transferred move e3g5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bg5+ (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



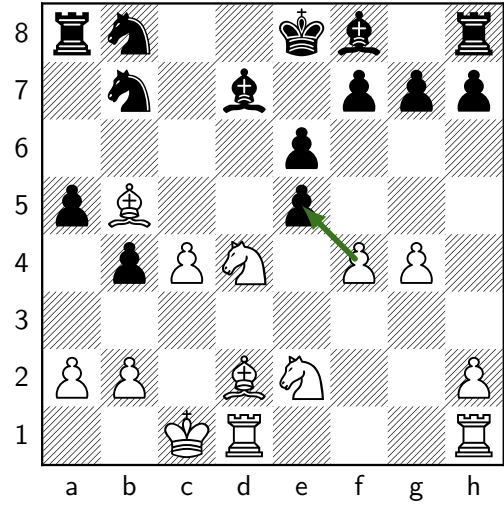
(a)



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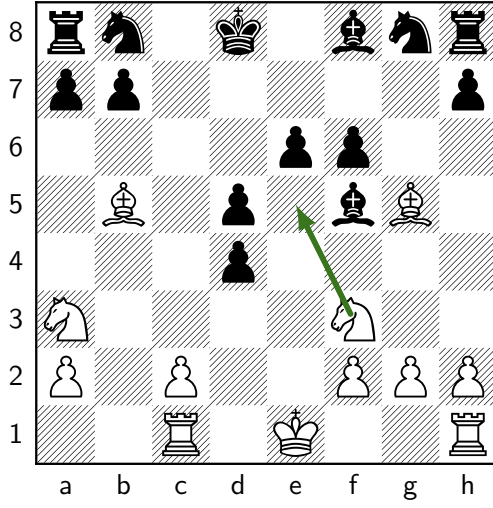


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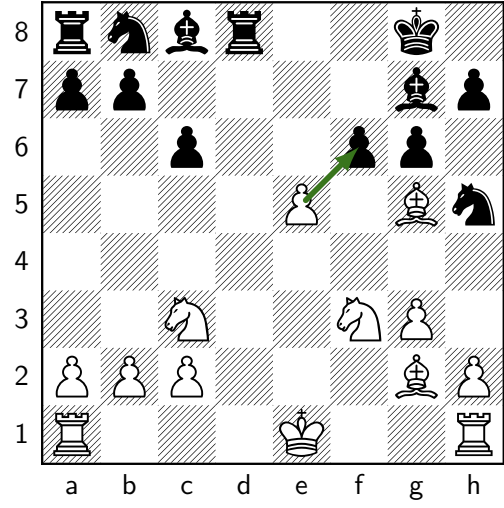


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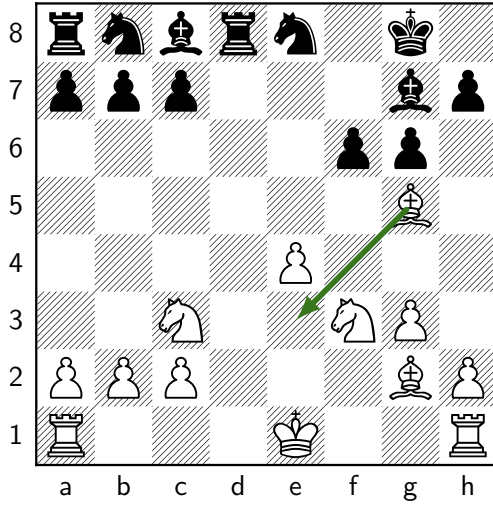
Figure 6: **Move 12, White: exf6.** (a) Our position; Melanie plays **exf6** (**e5f6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3535` (ply 24) [structural distance  $d=112$ , QAP obj=92]: the master played **exf6**; under  $\mu$  our *Pe5* plays the role of that game’s *Pe5*; cross-type roles: our *Pg2* as their *Bg2*, our *nb8* as their *Ph2*; transferred move **e5f6** (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#303` (ply 22) [structural distance  $d=116$ , QAP obj=73]: the master played **Nxd4**; under  $\mu$  our *Pe5* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Nf3* as their *Pb2*, our *Ph2* as their *ra8*; transferred move **e5f6** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3835` (ply 32) [structural distance  $d=102$ , QAP obj=71]: the master played **fxe5**; under  $\mu$  our *Pe5* plays the role of that game’s *Pf4*; cross-type roles: our *Nf3* as their *Rd1*, our *Pc2* as their *Rh1*; transferred move **e5f6** (**endorses this move**). Synthesis: 7 retrieved analogs transferred to **exf6** (summed QAP objective 361, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



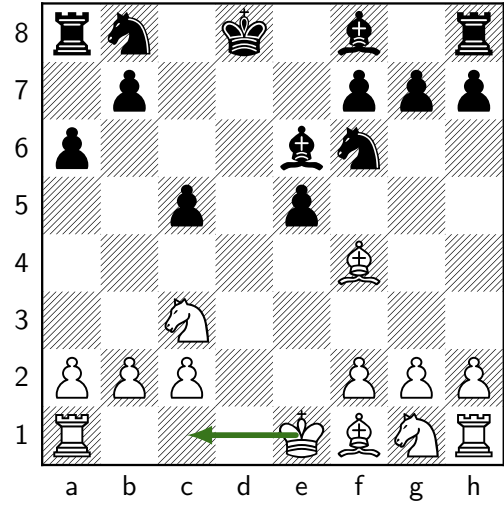
(a)



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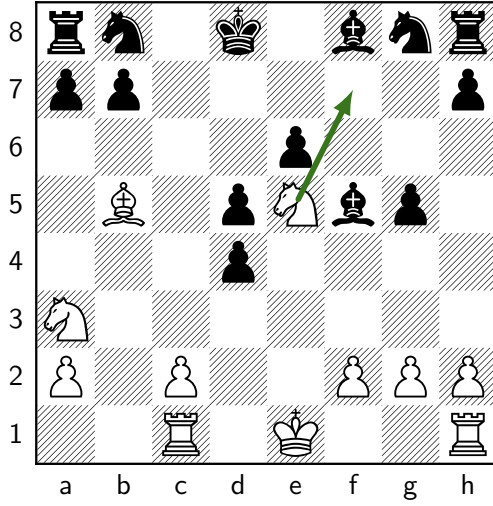
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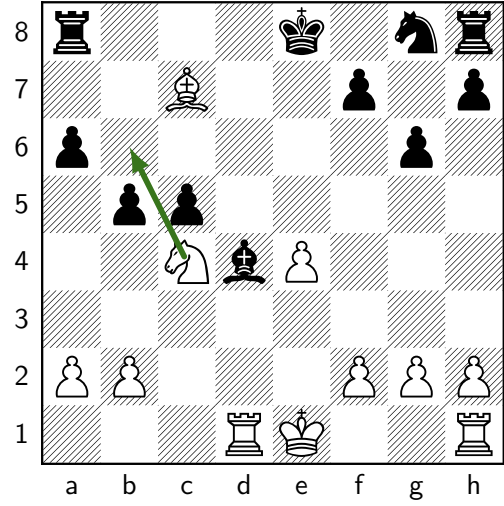
(d)

Figure 7: **Move 13, White: Ne5.** (a) Our position; Melanie plays Ne5 (f3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3535` (ply 24) [structural distance  $d=85$ , QAP obj=64]: the master played `exf6`; under  $\mu$  our `Bg5` plays the role of that game’s `Pe5` (a **cross-type** correspondence); cross-type roles: our `Na3` as their `Pc2`, our `pd4` as their `Bg2`; transferred move `g5h6`. (c) Retrieved from `ccrl_top_p0002#3534` (ply 22) [structural distance  $d=97$ , QAP obj=59]: the master played `Be3`; under  $\mu$  our `Bg5` plays the role of that game’s `Bg5`; cross-type roles: our `Na3` as their `Pb2`, our `Pa2` as their `Bg2`; transferred move `g5f4`. (d) Retrieved from `ccrl_top_p0001#274` (ply 16) [structural distance  $d=87$ , QAP obj=48]: the master played `O-O-O+`; under  $\mu$  our `Ke1` plays the role of that game’s `Ke1`; cross-type roles: our `pe6` as their `Ng1`, our `Pc2` as their `Rh1`; transferred move `e1g1`. Synthesis: no retrieved analog transferred to Ne5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

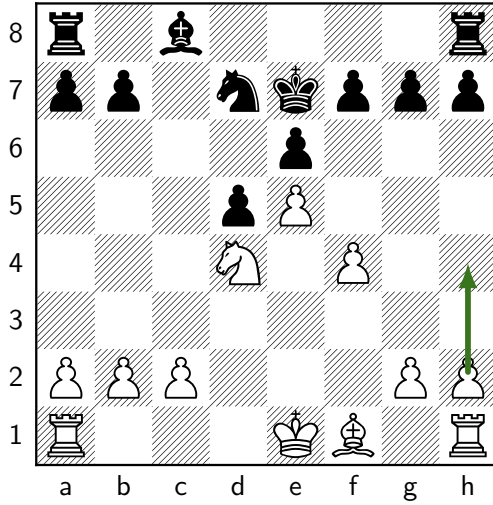




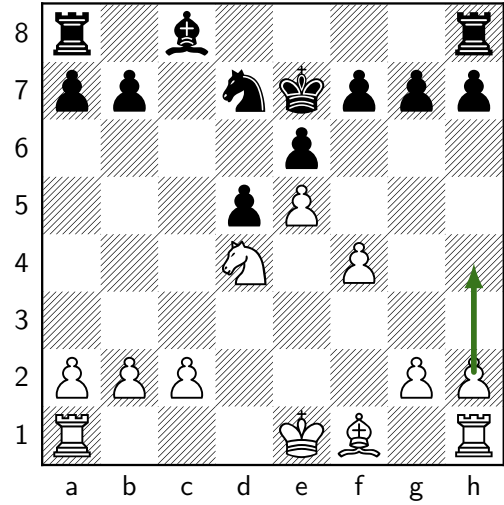
(a)



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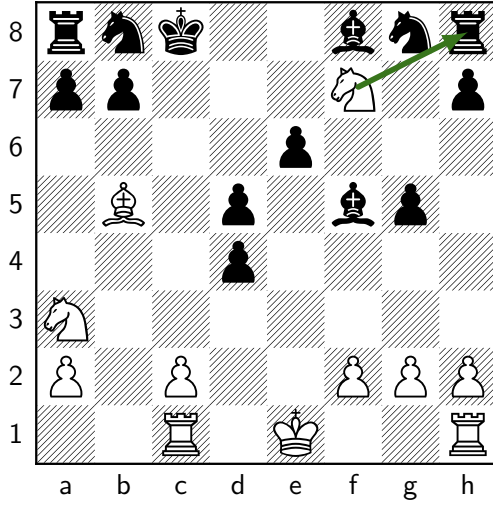


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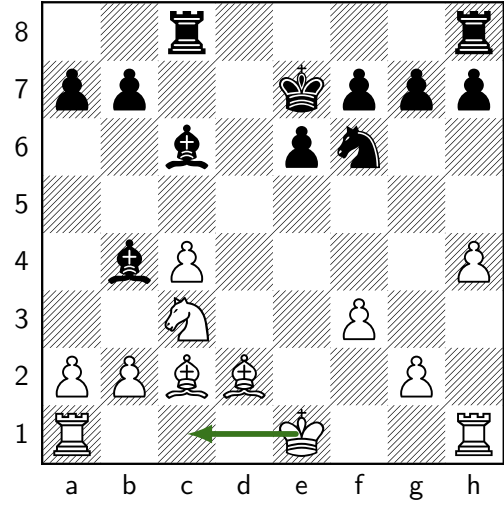


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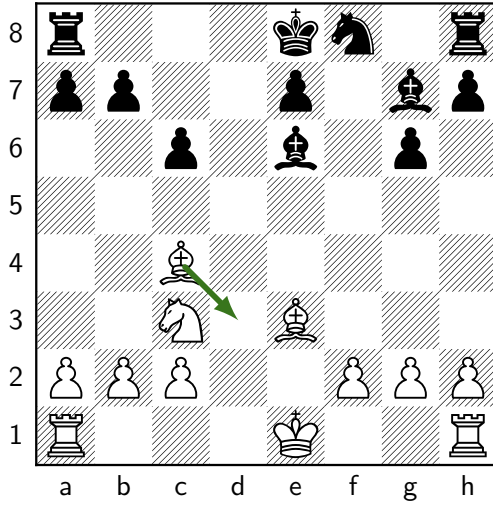
Figure 8: **Move 14, White: Nf7+.** (a) Our position; Melanie plays Nf7+ (e5f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#954` (ply 26) [structural distance  $d=85$ , QAP obj=4]: the master played Nb6; under  $\mu$  our Ne5 plays the role of that game’s Nc4; cross-type roles: our bf8 as their pb5, our Bb5 as their pg6; transferred move e5f7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#831` (ply 26) [structural distance  $d=85$ , QAP obj=29]: the master played h4; under  $\mu$  our Pg2 plays the role of that game’s Ph2; cross-type roles: our bf5 as their Nd4, our ra8 as their pd5; transferred move g2g4. (d) Retrieved from `ccrl_top_p0003#638` (ply 26) [structural distance  $d=85$ , QAP obj=29]: the master played h4; under  $\mu$  our Pg2 plays the role of that game’s Ph2; cross-type roles: our bf5 as their Nd4, our ra8 as their pd5; transferred move g2g4. Synthesis: 1 retrieved analog transferred to Nf7+ (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



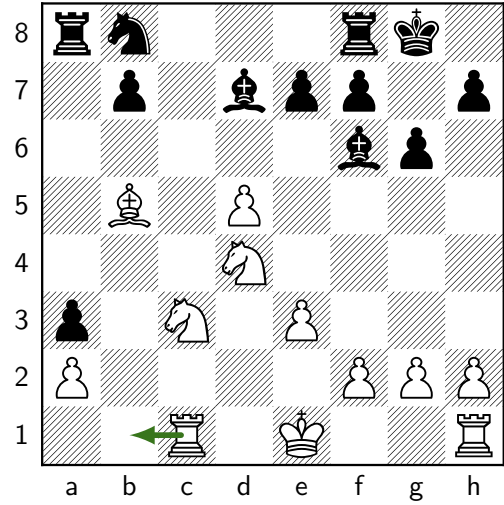
(a)



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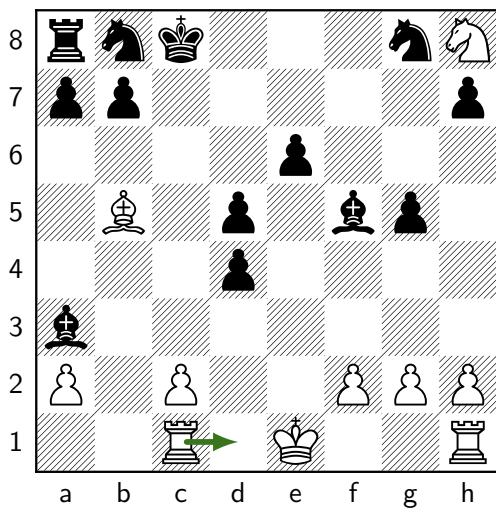


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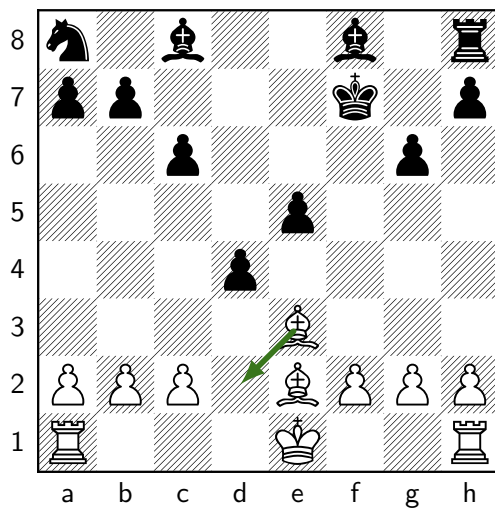


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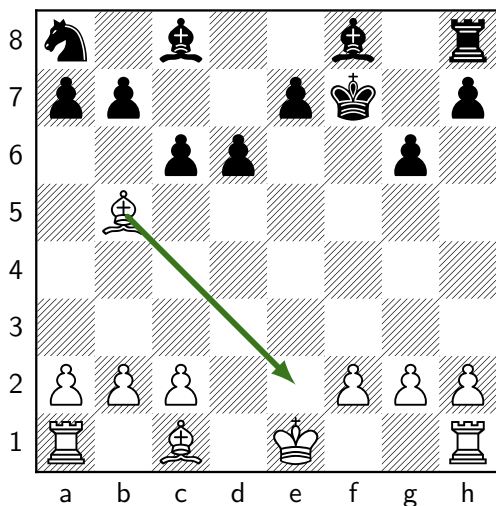
Figure 9: **Move 15, White: Nxh8.** (a) Our position; Melanie plays Nxh8 (f7h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#546` (ply 28) [structural distance  $d=74$ , QAP obj=20]: the master played 0-0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *ng8* as their *Pa2*, our *bf8* as their *Pb2*; transferred move e1g1. (c) Retrieved from `ccr1_top_p0001#821` (ply 26) [structural distance  $d=79$ , QAP obj=20]: the master played Bd3; under  $\mu$  our *Nf7* plays the role of that game’s *Bc4* (a **cross-type** correspondence); cross-type roles: our *rh8* as their *be6*, our *ng8* as their *pb7*; transferred move f7g5. (d) Retrieved from `ccr1_top_p0001#115` (ply 30) [structural distance  $d=70$ , QAP obj=19]: the master played Rb1; under  $\mu$  our *Rc1* plays the role of that game’s *Rc1*; cross-type roles: our *Bb5* as their *Nc3*, our *rh8* as their *Nd4*; transferred move c1b1. Synthesis: no retrieved analog transferred to Nxh8 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



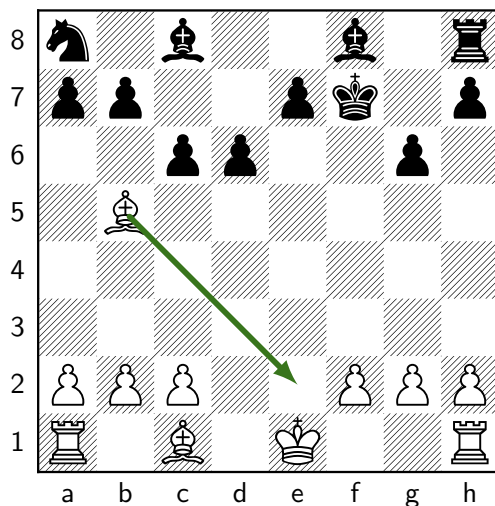
(a)



(b)

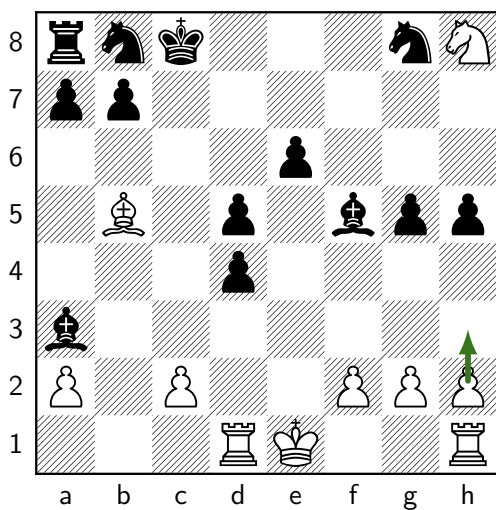


(c)

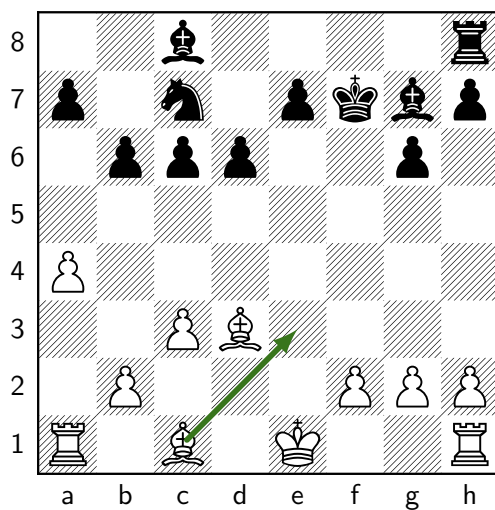


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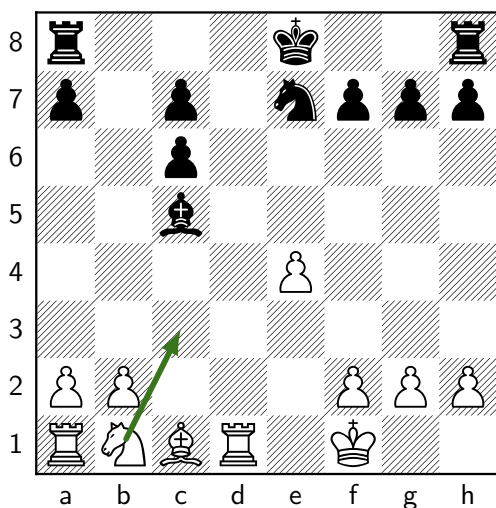
Figure 10: **Move 16, White: Rd1.** (a) Our position; Melanie plays Rd1 (c1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1329` (ply 40) [structural distance  $d=58$ , QAP obj=18]: the master played Bd2; under  $\mu$  our Rc1 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our ng8 as their Ra1, our Bb5 as their Pb2; transferred move c1d1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2696` (ply 30) [structural distance  $d=57$ , QAP obj=17]: the master played Be2; under  $\mu$  our Rc1 plays the role of that game’s Bb5 (a **cross-type** correspondence); cross-type roles: our Nh8 as their Ra1, our bf5 as their Pg2; transferred move c1d1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2827` (ply 34) [structural distance  $d=57$ , QAP obj=17]: the master played Be2; under  $\mu$  our Rc1 plays the role of that game’s Bb5 (a **cross-type** correspondence); cross-type roles: our Nh8 as their Ra1, our bf5 as their Pg2; transferred move c1d1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rd1 (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



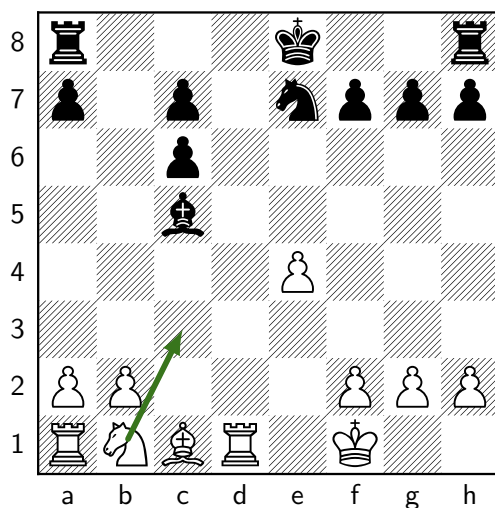
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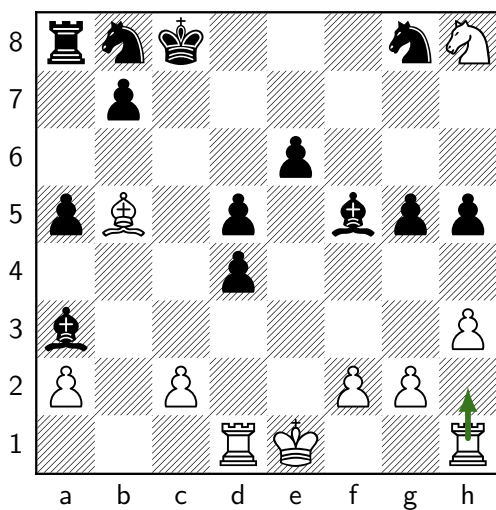


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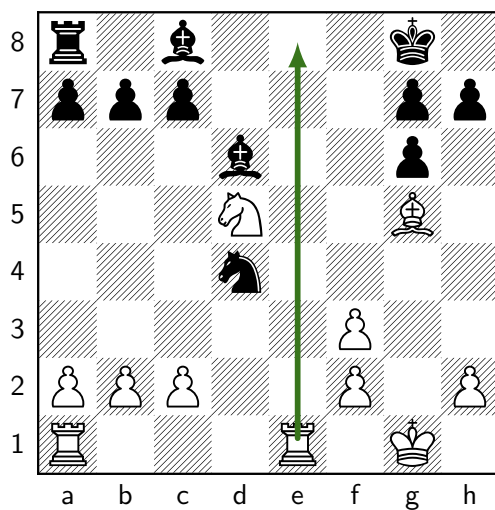


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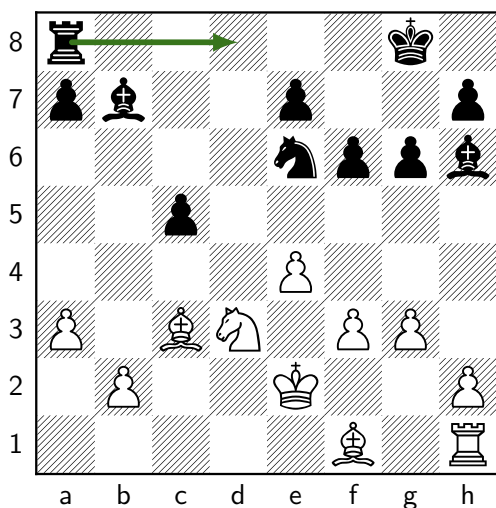
Figure 11: **Move 17, White: h3.** (a) Our position; Melanie plays h3 (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1328` (ply 36) [structural distance  $d=65$ , QAP obj=33]: the master played Be3; under  $\mu$  our *Ph2* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Pa2* as their *Rh1*, our *ng8* as their *Pg2*; transferred move h2h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1729` (ply 24) [structural distance  $d=69$ , QAP obj=21]: the master played Nc3; under  $\mu$  our *Ph2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *pe6* as their *Bc1*, our *Nh8* as their *Pa2*; transferred move h2h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1730` (ply 24) [structural distance  $d=69$ , QAP obj=21]: the master played Nc3; under  $\mu$  our *Ph2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *pe6* as their *Bc1*, our *Nh8* as their *Pa2*; transferred move h2h3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to h3 (summed QAP objective 131, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



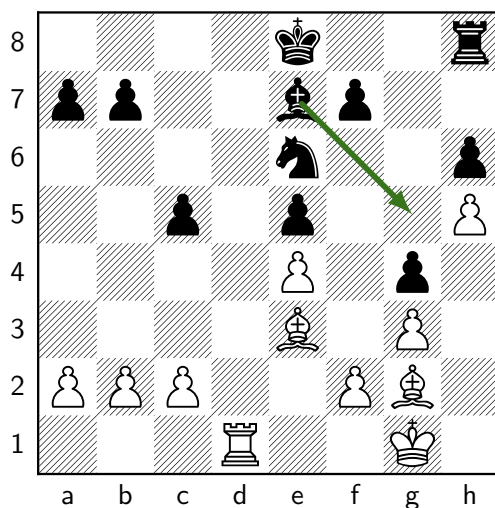
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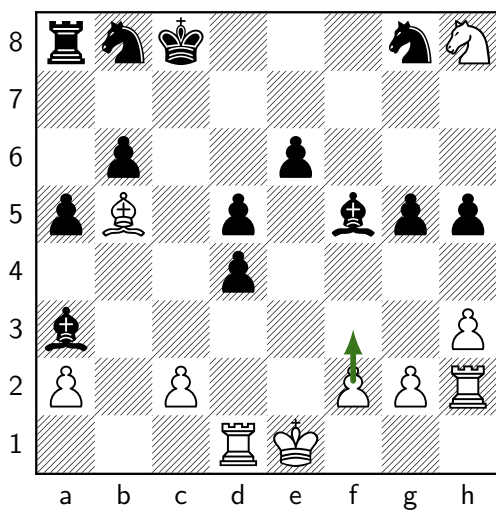


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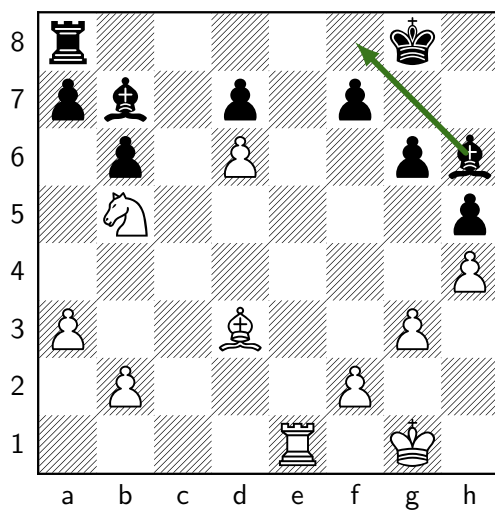


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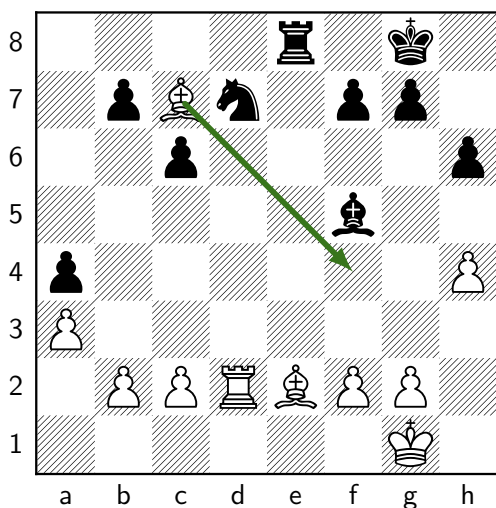
Figure 12: **Move 18, White: Rh2.** (a) Our position; Melanie plays Rh2 (**h1h2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1842` (ply 30) [structural distance  $d=55$ , QAP obj=52]: the master played **Re8+**; under  $\mu$  our *Rh1* plays the role of that game’s *Re1*; cross-type roles: our *ba3* as their *Ra1*, our *Nh8* as their *Pa2*; transferred move **h1h2** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2273` (ply 37) [structural distance  $d=73$ , QAP obj=41]: the master played **Rd8**; under  $\mu$  our *Rh1* plays the role of that game’s *Ra1*; cross-type roles: our *Rd1* as their *bc3*, our *bf5* as their *nd3*; transferred move **h1h2** (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2869` (ply 43) [structural distance  $d=71$ , QAP obj=36]: the master played **Bg5**; under  $\mu$  our *Rh1* plays the role of that game’s *Be2* (a **cross-type** correspondence); cross-type roles: our *ba3* as their *Pa7*, our *ra8* as their *Pb7*; transferred move **h1h2** (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Rh2 (summed QAP objective 215, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



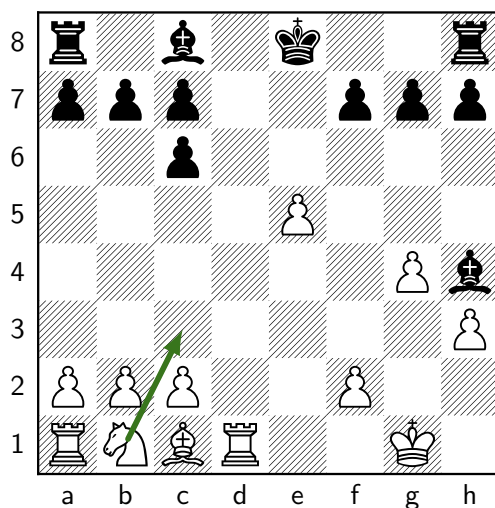
(a)



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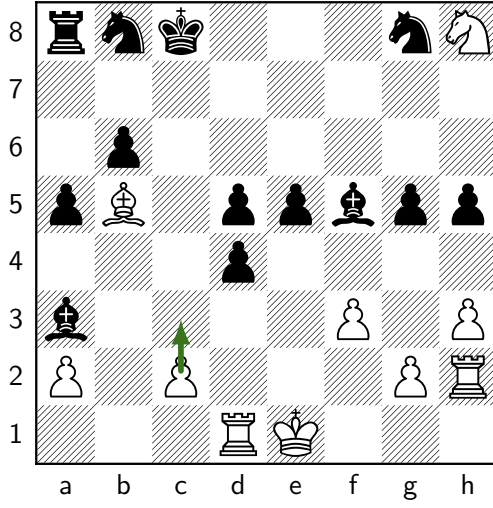


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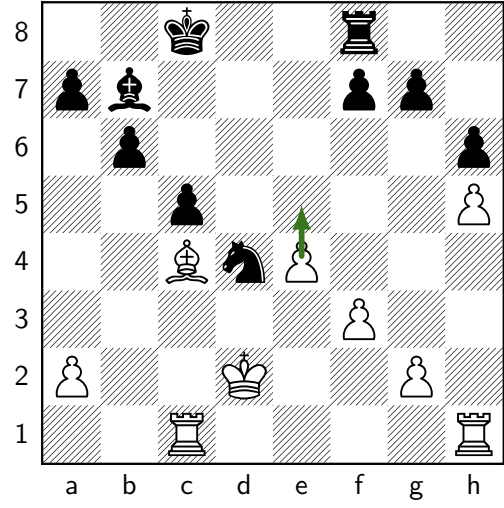


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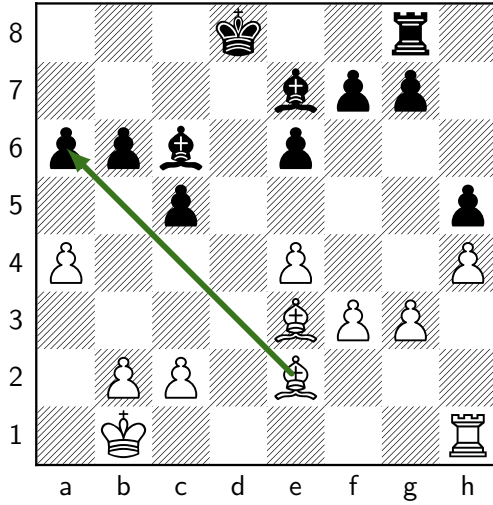
Figure 13: **Move 19, White: f3**. (a) Our position; Melanie plays f3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1164` (ply 49) [structural distance  $d=66$ , QAP obj=35]: the master played Bf8; under  $\mu$  our Pf2 plays the role of that game’s Bh3 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Bb7, our bf5 as their nb5; transferred move f2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#265` (ply 44) [structural distance  $d=50$ , QAP obj=33]: the master played Bf4; under  $\mu$  our Pf2 plays the role of that game’s Bc7 (a **cross-type** correspondence); cross-type roles: our Bb5 as their Pb2, our Ph3 as their Be2; transferred move f2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2255` (ply 24) [structural distance  $d=68$ , QAP obj=28]: the master played Nc3; under  $\mu$  our Pf2 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our pa5 as their Ra1, our nb8 as their Bc1; transferred move f2f3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to f3 (summed QAP objective 121, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



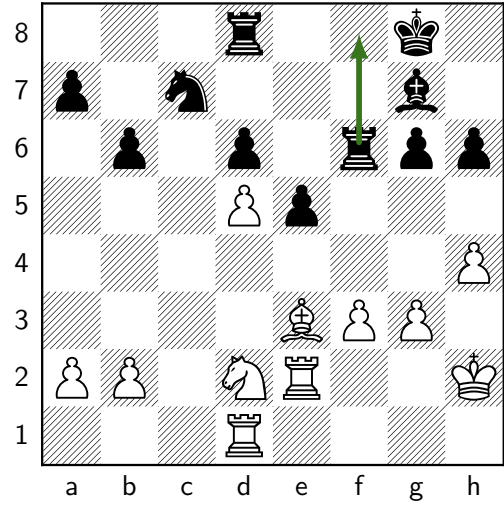
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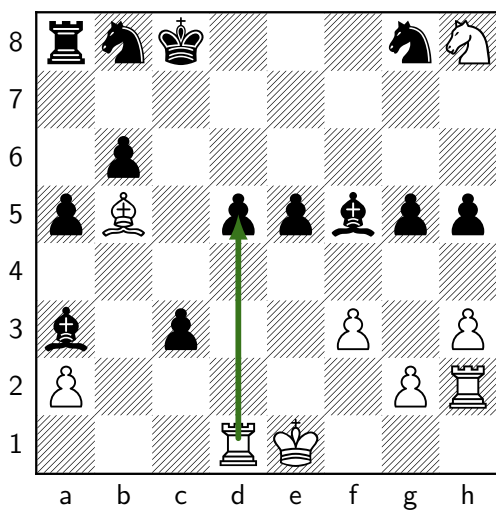
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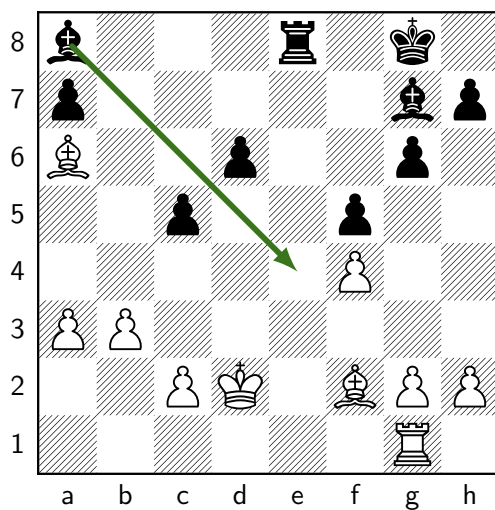
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Figure 14: **Move 20, White: c3.** (a) Our position; Melanie plays c3 (c2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0000#348` (ply 44) [structural distance  $d=72$ , QAP obj=42]: the master played e5; under  $\mu$  our  $Pc2$  plays the role of that game’s  $Pe4$ ; cross-type roles: our  $nb8$  as their  $Rc1$ , our  $Pa2$  as their  $Rh1$ ; transferred move c2c3 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#121` (ply 50) [structural distance  $d=64$ , QAP obj=57]: the master played Bxa6; under  $\mu$  our  $Pf3$  plays the role of that game’s  $Be2$  (a **cross-type** correspondence); cross-type roles: our  $Bb5$  as their  $Rh1$ , our  $Rd1$  as their  $be7$ ; transferred move f3f4. (d) Retrieved from `ccl1_top_p0001#1267` (ply 53) [structural distance  $d=59$ , QAP obj=49]: the master played Rff8; under  $\mu$  our  $Rd1$  plays the role of that game’s  $Rf3$ ; cross-type roles: our  $nb8$  as their  $Rd8$ , our  $Nh8$  as their  $Pa7$ ; transferred move d1d2. Synthesis: 1 retrieved analog transferred to c3 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

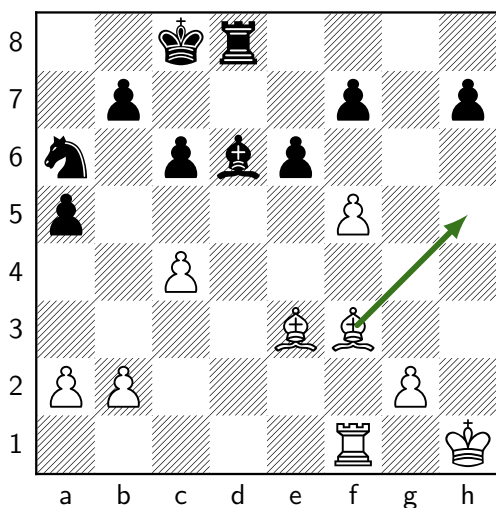




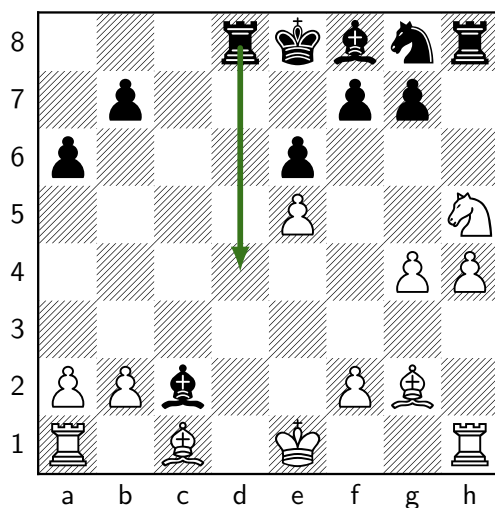
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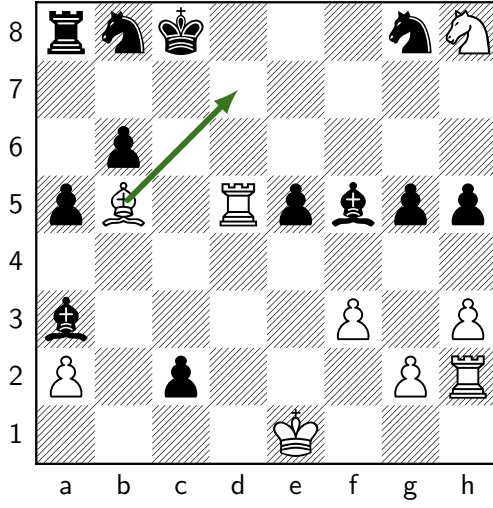
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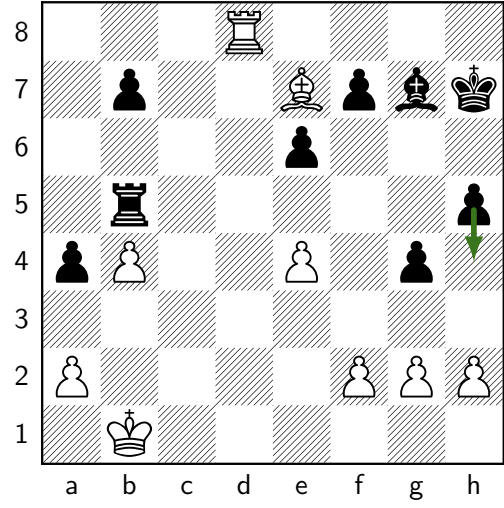
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Figure 15: **Move 21, White: Rxd5**. (a) Our position; Melanie plays Rxd5 (d1d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3390` (ply 43) [structural distance  $d=60$ , QAP obj=21]: the master played Be4; under  $\mu$  our *Rd1* plays the role of that game’s *Ba1* (a **cross-type** correspondence); cross-type roles: our *Pf3* as their *Re8*, our *pg5* as their *Bg7*; transferred move d1d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3272` (ply 48) [structural distance  $d=66$ , QAP obj=17]: the master played Bh5; under  $\mu$  our *Rd1* plays the role of that game’s *Bf3* (a **cross-type** correspondence); cross-type roles: our *Pf3* as their *Be3*, our *Nh8* as their *Pc4*; transferred move d1d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#934` (ply 29) [structural distance  $d=69$ , QAP obj=13]: the master played Rd4; under  $\mu$  our *Rd1* plays the role of that game’s *Rd1*; cross-type roles: our *pg5* as their *Bf8*, our *ph5* as their *Ng8*; transferred move d1d5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rxd5 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

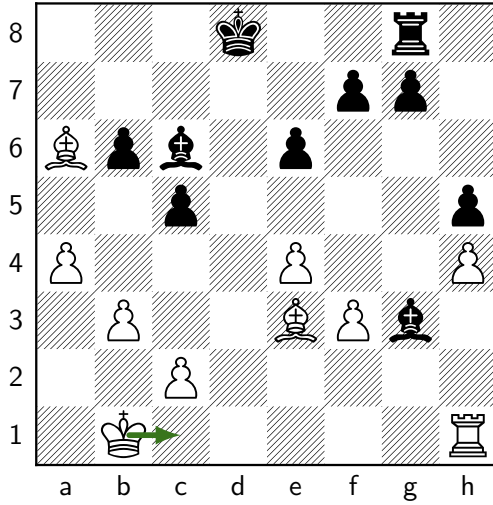




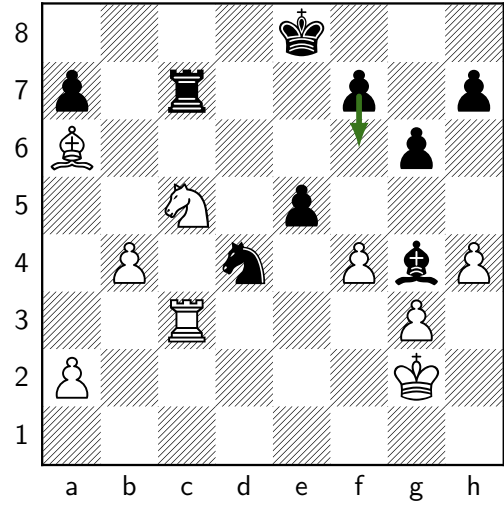
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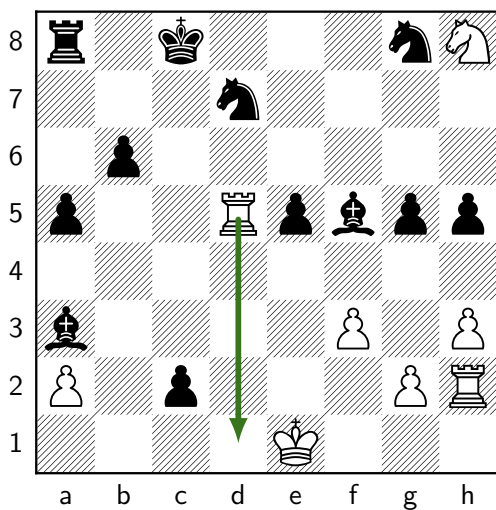


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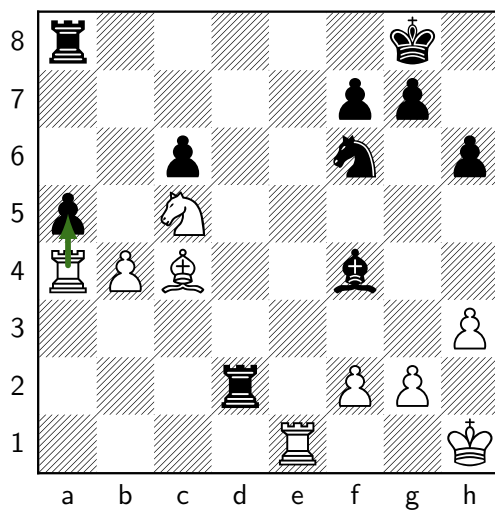


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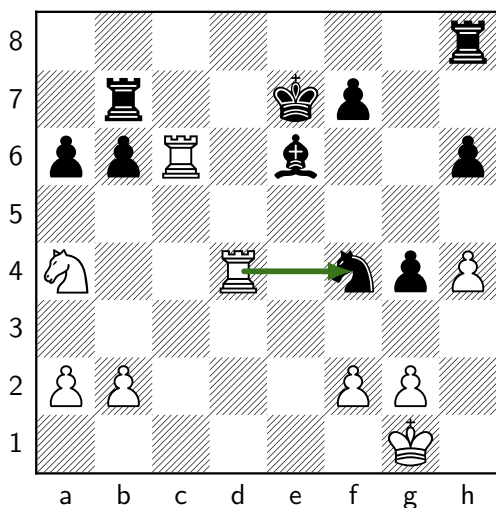
Figure 16: **Move 22, White: Bd7+.** (a) Our position; Melanie plays Bd7+ (b5d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2149` (ply 55) [structural distance  $d=63$ , QAP obj=43]: the master played h4; under  $\mu$  our *Nh8* plays the role of that game’s *Ph4* (a **cross-type** correspondence); cross-type roles: our *pg5* as their *rd8*, our *Bb5* as their *Pb7*; transferred move h8g6. (c) Retrieved from `ccr1_top_p0000#121` (ply 54) [structural distance  $d=69$ , QAP obj=36]: the master played Kc1; under  $\mu$  our *Ke1* plays the role of that game’s *Kb1*; cross-type roles: our *pb6* as their *Rh1*, our *Rh2* as their *Pc2*; transferred move e1f1. (d) Retrieved from `ccr1_top_p0003#33` (ply 55) [structural distance  $d=70$ , QAP obj=34]: the master played f6; under  $\mu$  our *Pf3* plays the role of that game’s *Pf2*; cross-type roles: our *Bb5* as their *Pa7*, our *pe5* as their *nc5*; transferred move f3f4. Synthesis: no retrieved analog transferred to Bd7+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



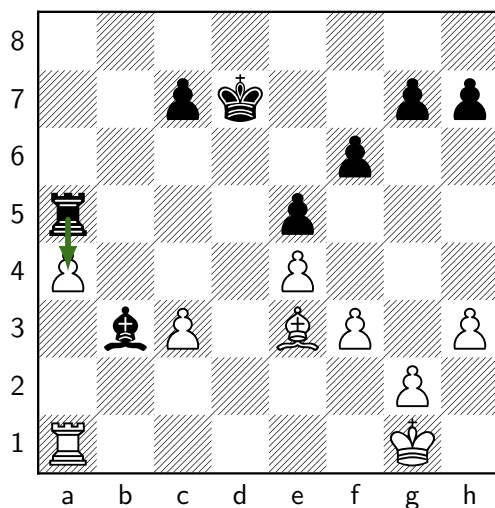
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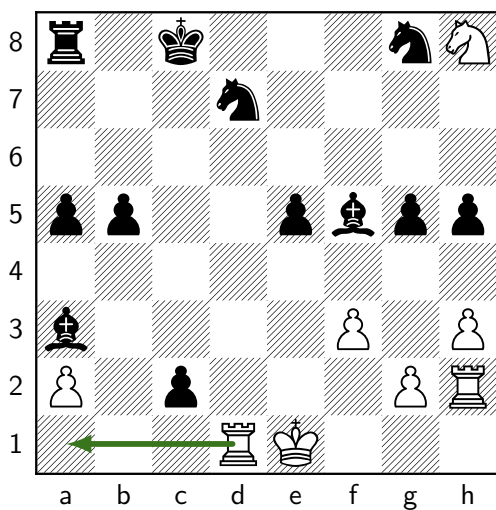


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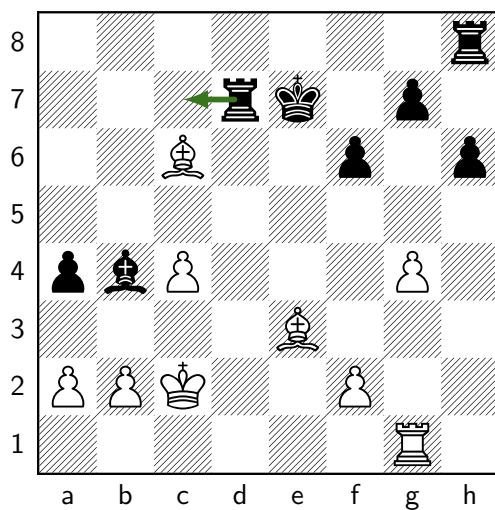


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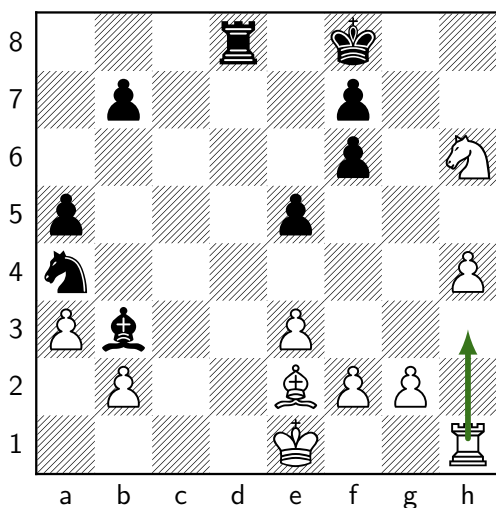
Figure 17: **Move 23, White: Rd1.** (a) Our position; Melanie plays Rd1 (d5d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2143` (ply 54) [structural distance  $d=70$ , QAP obj=53]: the master played  $Rxa5$ ; under  $\mu$  our  $Rd5$  plays the role of that game’s  $Ra4$ ; cross-type roles: our  $pc2$  as their  $Re1$ , our  $bf5$  as their  $rd2$ ; transferred move  $d5e5$ . (c) Retrieved from `ccr1_top_p0002#2688` (ply 58) [structural distance  $d=68$ , QAP obj=50]: the master played  $Rxf4$ ; under  $\mu$  our  $Rd5$  plays the role of that game’s  $Rd4$ ; cross-type roles: our  $Rd5$  as their  $Na4$ , our  $bf5$  as their  $Ph4$ ; transferred move  $d5e5$ . (d) Retrieved from `ccr1_top_p0003#1348` (ply 61) [structural distance  $d=68$ , QAP obj=46]: the master played  $Rxa4$ ; under  $\mu$  our  $Rd5$  plays the role of that game’s  $Ra4$ ; cross-type roles: our  $Rh2$  as their  $Pg7$ , our  $Rd5$  as their  $Bb3$ ; transferred move  $d5e5$ . Synthesis: no retrieved analog transferred to Rd1 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



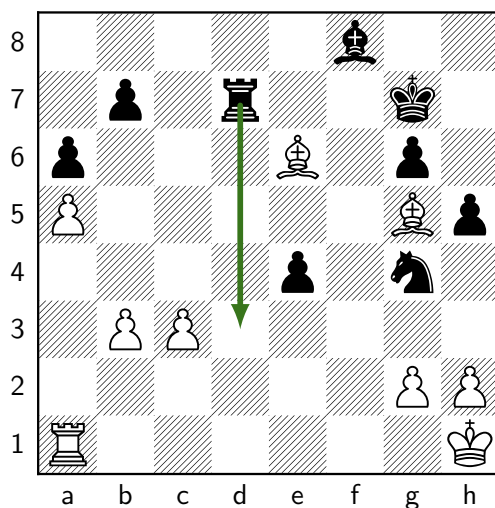
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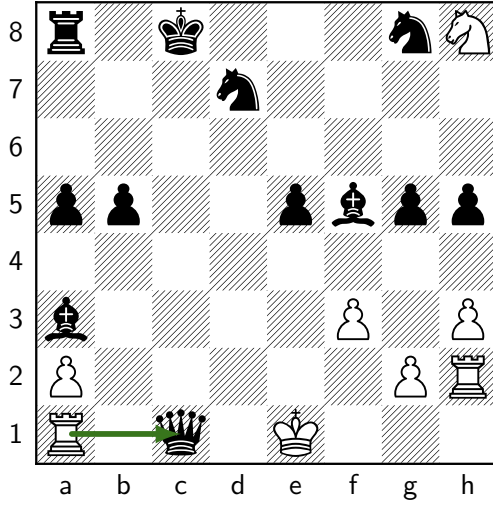


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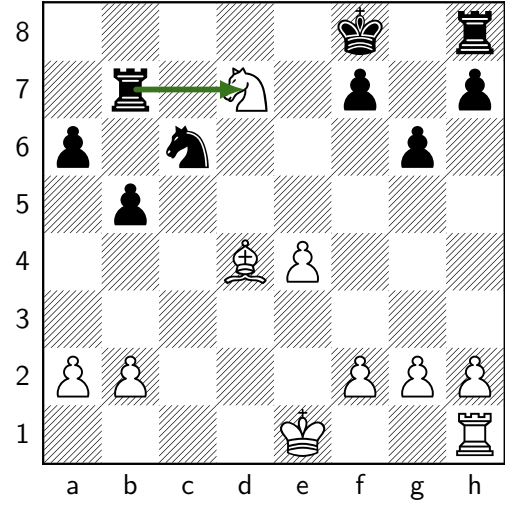


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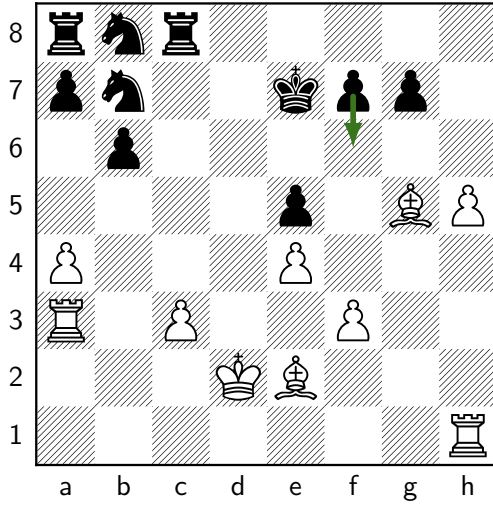
Figure 18: **Move 24, White: Ra1.** (a) Our position; Melanie plays **Ra1** (**d1a1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1423` (ply 53) [structural distance  $d=56$ , QAP obj=26]: the master played **Rc7**; under  $\mu$  our *Rd1* plays the role of that game’s *Rd2*; cross-type roles: our *pc2* as their *bc6*, our *Nh8* as their *Bb4*; transferred move **d1a1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#380` (ply 50) [structural distance  $d=56$ , QAP obj=4]: the master played **Rh3**; under  $\mu$  our *Rd1* plays the role of that game’s *Rh1*; cross-type roles: our *pa5* as their *Be2*, our *Rh2* as their *Pg2*; transferred move **d1a1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#190` (ply 57) [structural distance  $d=58$ , QAP obj=31]: the master played **Rd3**; under  $\mu$  our *Rd1* plays the role of that game’s *Rd2*; cross-type roles: our *Nh8* as their *Bf8*, our *pc2* as their *be6*; transferred move **d1d5**. Synthesis: 2 retrieved analogs transferred to **Ra1** (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



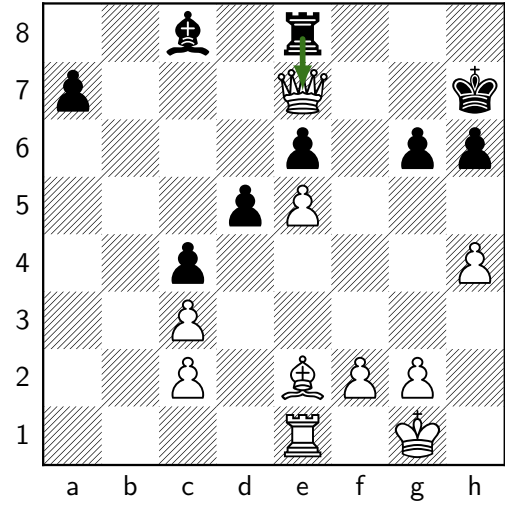
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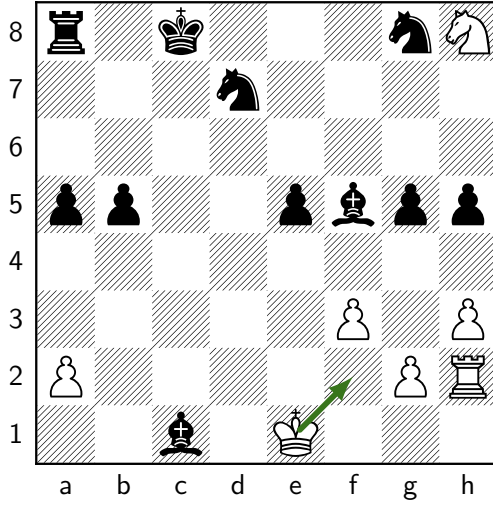


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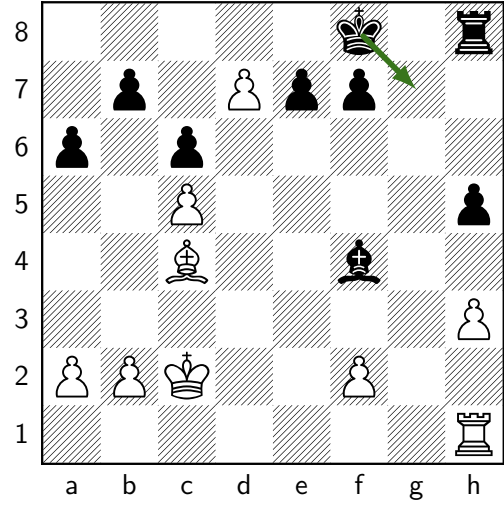


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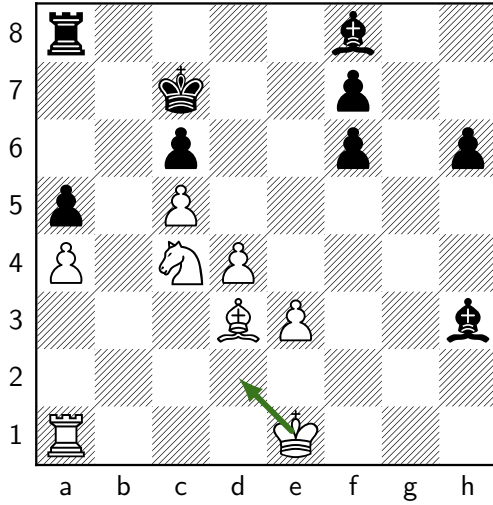
Figure 19: **Move 25, White: Rxc1+.** (a) Our position; Melanie plays Rxc1+ (a1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#954` (ply 39) [structural distance  $d=84$ , QAP obj=73]: the master played Rxd7; under  $\mu$  our *Ra1* plays the role of that game’s *Rb2*; cross-type roles: our *qc1* as their *nd7*, our *Nh8* as their *Ph7*; transferred move a1c1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3575` (ply 55) [structural distance  $d=92$ , QAP obj=68]: the master played f6; under  $\mu$  our *Ra1* plays the role of that game’s *Pf2* (a **cross-type** correspondence); cross-type roles: our *Nh8* as their *Ra8*, our *pg5* as their *Nb8*; transferred move a1c1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1952` (ply 65) [structural distance  $d=91$ , QAP obj=49]: the master played Rxe7; under  $\mu$  our *Ra1* plays the role of that game’s *Re1*; cross-type roles: our *pa5* as their *Bc8*, our *ng8* as their *pe5*; transferred move a1c1 (**endorses this move**). Synthesis: 21 retrieved analogs transferred to Rxc1+ (summed QAP objective 324, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



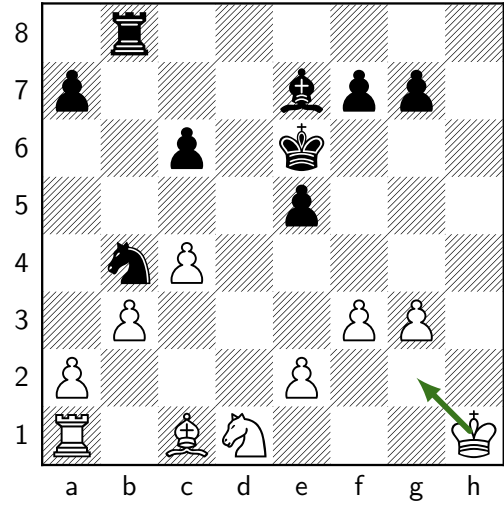
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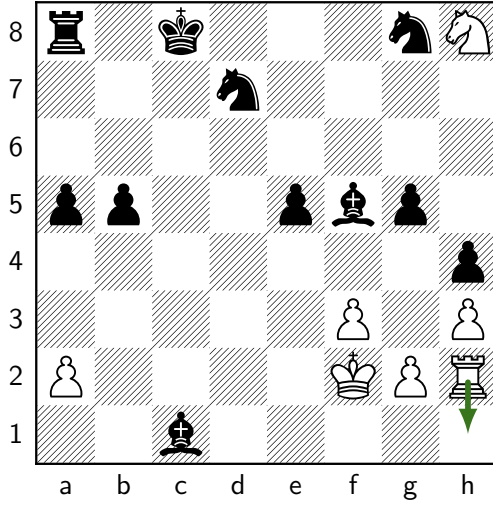


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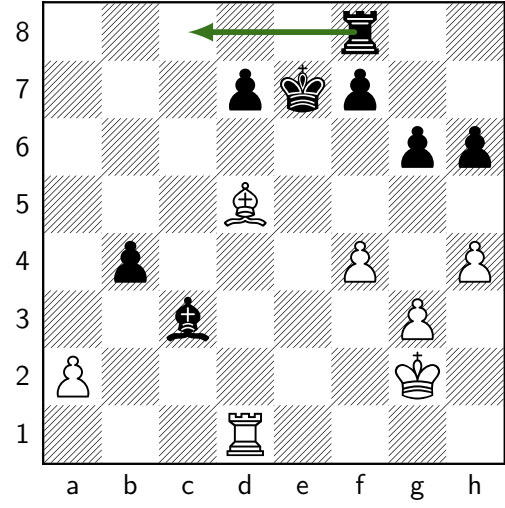


(d)

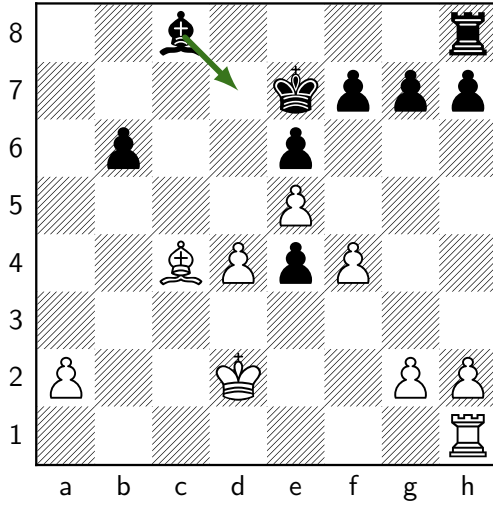
Figure 20: **Move 26, White: Kf2.** (a) Our position; Melanie plays Kf2 (e1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1654` (ply 53) [structural distance  $d=45$ , QAP obj=31]: the master played Kg7; under  $\mu$  our *Ke1* plays the role of that game’s *Kf1*; cross-type roles: our *ph5* as their *Rh8*, our *Rh2* as their *Pb7*; transferred move e1f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#115` (ply 50) [structural distance  $d=48$ , QAP obj=24]: the master played Kd2; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pg5* as their *Ra1*, our *Pf3* as their *Bd3*; transferred move e1f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1498` (ply 46) [structural distance  $d=39$ , QAP obj=23]: the master played Kg2; under  $\mu$  our *Ke1* plays the role of that game’s *Kh1*; cross-type roles: our *Pa2* as their *Nd1*, our *bf5* as their *nb4*; transferred move e1f2 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kf2 (summed QAP objective 190, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



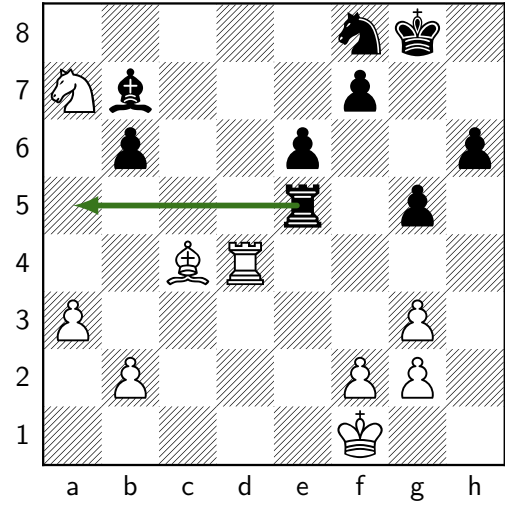
(a)



(b)

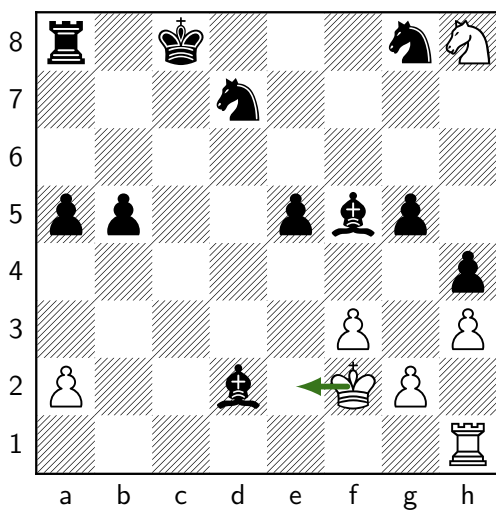


(c)

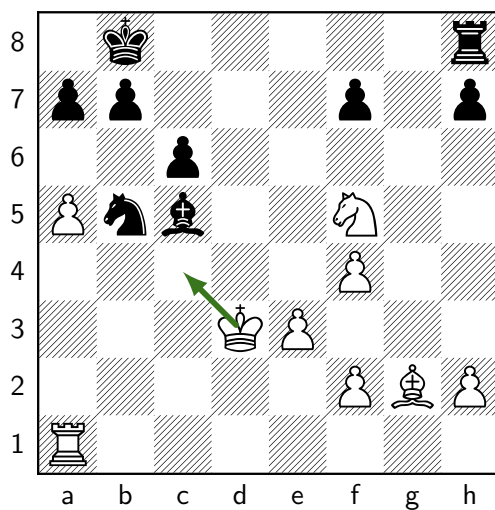


(d)

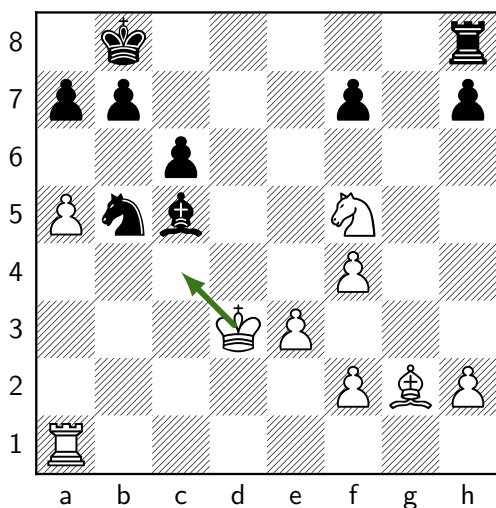
Figure 21: **Move 27, White: Rh1.** (a) Our position; Melanie plays Rh1 (**h2h1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#98` (ply 55) [structural distance  $d=50$ , QAP obj=23]: the master played Rc8; under  $\mu$  our Rh2 plays the role of that game’s Rf1; cross-type roles: our Nh8 as their Pb4, our bc1 as their ph4; transferred move **h2h1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0003#1750` (ply 49) [structural distance  $d=46$ , QAP obj=23]: the master played Bd7; under  $\mu$  our Rh2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our bc1 as their Rh8, our ng8 as their Ph7; transferred move **h2h1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0003#2920` (ply 55) [structural distance  $d=49$ , QAP obj=21]: the master played Ra5; under  $\mu$  our Rh2 plays the role of that game’s Re4; cross-type roles: our ph4 as their Nf8, our Pf3 as their na7; transferred move **h2h1 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to Rh1 (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



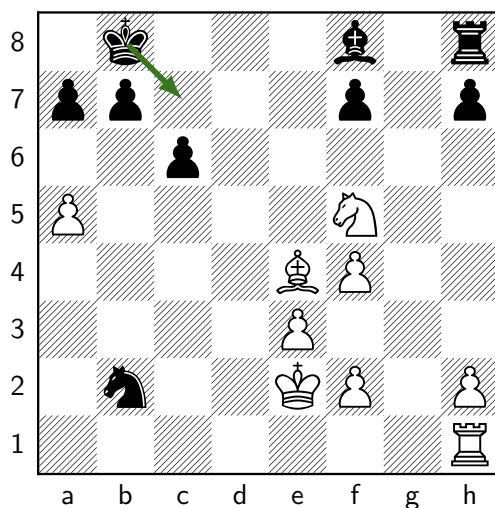
(a)



(b)

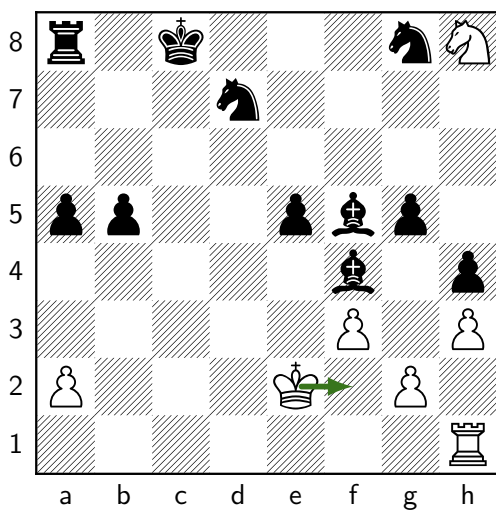


(c)

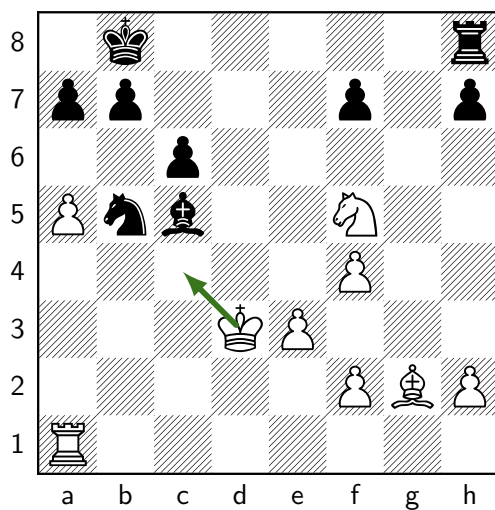


(d)

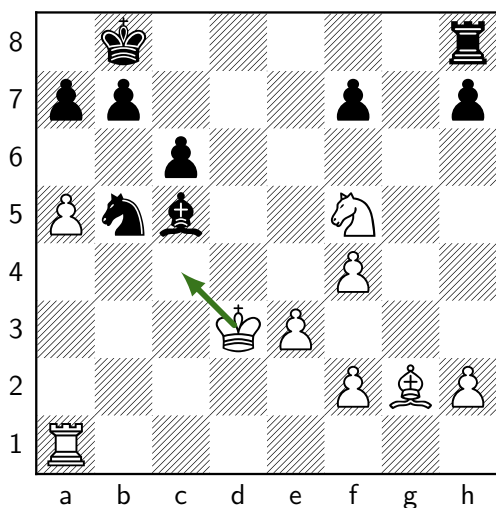
Figure 22: **Move 28, White: Ke2.** (a) Our position; Melanie plays Ke2 (f2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1451` (ply 52) [structural distance  $d=47$ , QAP obj=24]: the master played Kc4; under  $\mu$  our Kf2 plays the role of that game's Kd3; cross-type roles: our bd2 as their nb5, our nd7 as their pa7; transferred move f2e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2418` (ply 52) [structural distance  $d=47$ , QAP obj=24]: the master played Kc4; under  $\mu$  our Kf2 plays the role of that game's Kd3; cross-type roles: our bd2 as their nb5, our nd7 as their pa7; transferred move f2e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2456` (ply 47) [structural distance  $d=49$ , QAP obj=22]: the master played Kc7; under  $\mu$  our Kf2 plays the role of that game's Kb1; cross-type roles: our Pa2 as their Bf8, our bd2 as their Rh8; transferred move f2e2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ke2 (summed QAP objective 91, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



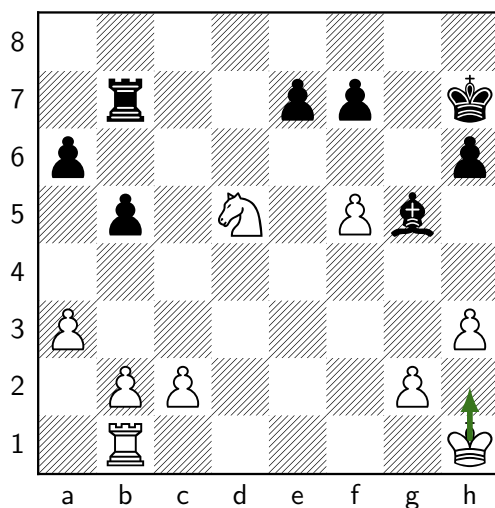
(a)



(b)



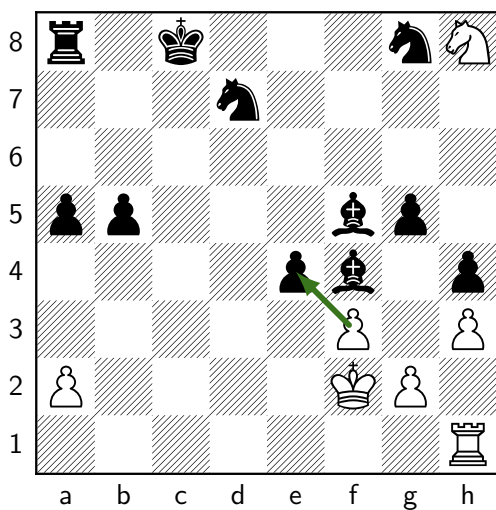
(c)



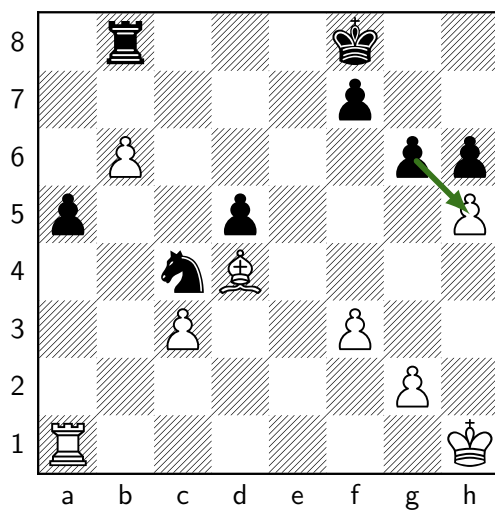
(d)

Figure 23: **Move 29, White: Kf2.** (a) Our position; Melanie plays Kf2 (e2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1451` (ply 52) [structural distance  $d=52$ , QAP obj=24]: the master played Kc4; under  $\mu$  our Ke2 plays the role of that game’s Kd3; cross-type roles: our pg5 as their Ra1, our bf4 as their Pf4; transferred move e2f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2418` (ply 52) [structural distance  $d=52$ , QAP obj=24]: the master played Kc4; under  $\mu$  our Ke2 plays the role of that game’s Kd3; cross-type roles: our pg5 as their Ra1, our bf4 as their Pf4; transferred move e2f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1988` (ply 58) [structural distance  $d=48$ , QAP obj=22]: the master played Kh2; under  $\mu$  our Ke2 plays the role of that game’s Kh1; cross-type roles: our pe5 as their Rb1, our ra8 as their Pc2; transferred move e2f2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kf2 (summed QAP objective 97, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

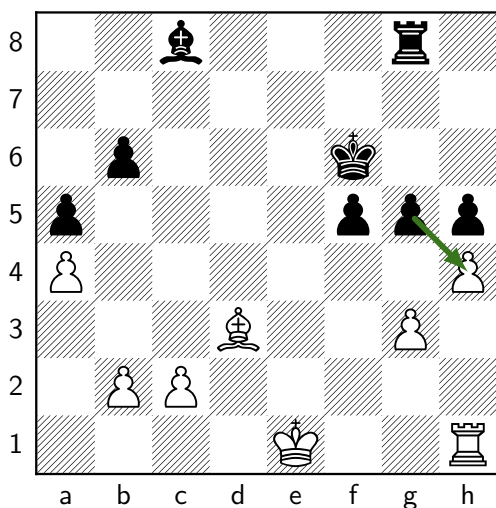




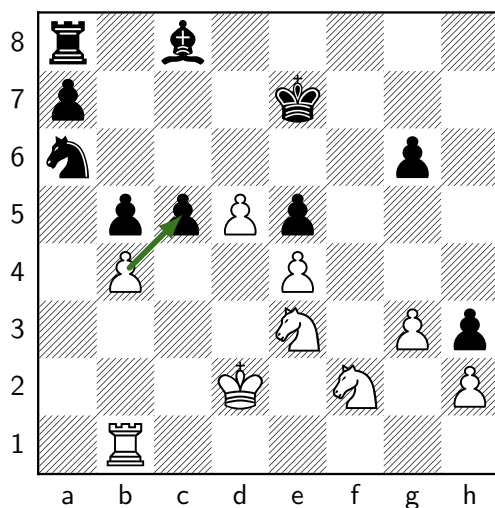
(a)



(b)

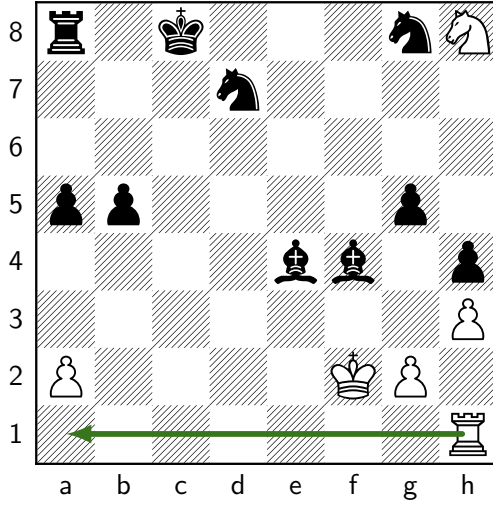


(c)

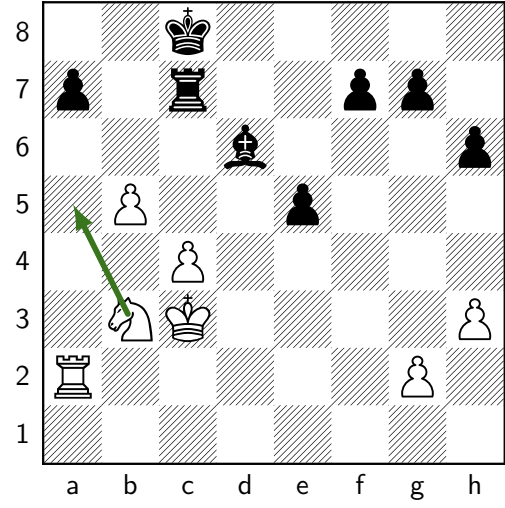


(d)

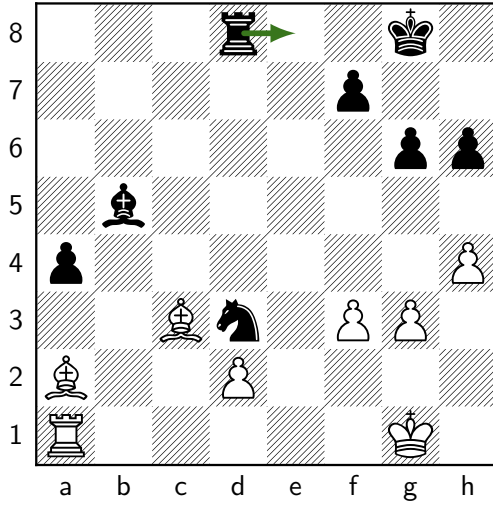
Figure 24: **Move 30, White: fxe4**. (a) Our position; Melanie plays fxe4 (f3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#3858 (ply 75) [structural distance  $d=54$ , QAP obj=54]: the master played gxh5; under  $\mu$  our Pf3 plays the role of that game’s Pg3; cross-type roles: our bf5 as their Rb8, our bf4 as their Ph6; transferred move f3e4 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0002#2474 (ply 67) [structural distance  $d=49$ , QAP obj=53]: the master played gxh4; under  $\mu$  our Pf3 plays the role of that game’s Pg4; cross-type roles: our Nh8 as their Bc8, our Pg2 as their Rg8; transferred move f3e4 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0000#3720 (ply 68) [structural distance  $d=52$ , QAP obj=53]: the master played bxc5; under  $\mu$  our Pf3 plays the role of that game’s Pb4; cross-type roles: our ng8 as their Rb1, our bf5 as their Nf2; transferred move f3e4 (**endorses this move**). Synthesis: 17 retrieved analogs transferred to fxe4 (summed QAP objective 439, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



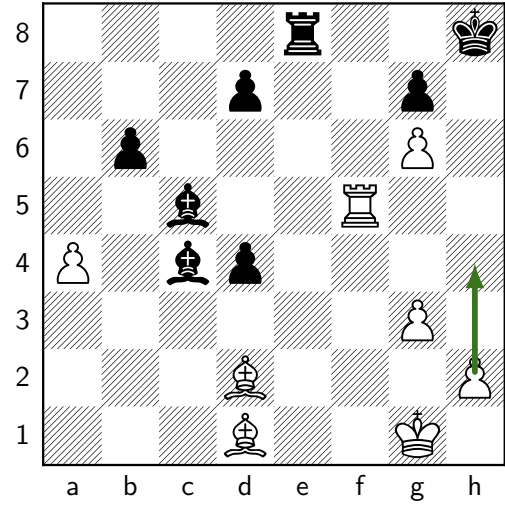
(a)



(b)

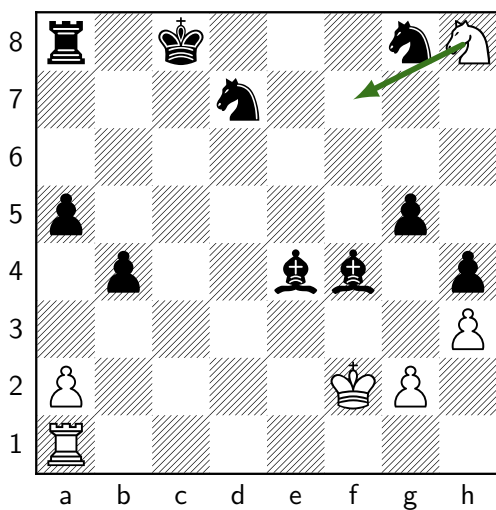


(c)

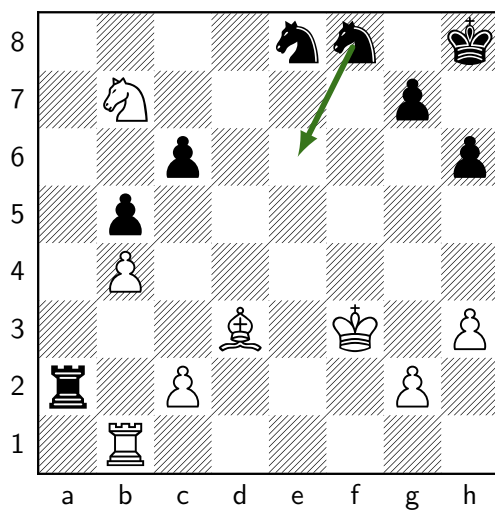


(d)

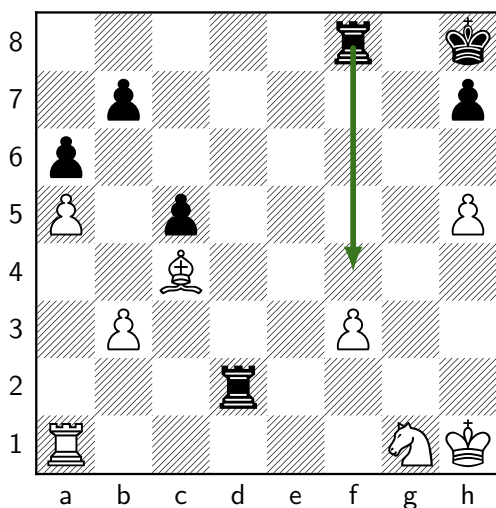
Figure 25: **Move 31, White: Ra1**. (a) Our position; Melanie plays Ra1 (h1a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0003#745 (ply 76) [structural distance  $d=38$ , QAP obj=32]: the master played Na5; under  $\mu$  our Rh1 plays the role of that game's Nb3 (a **cross-type** correspondence); cross-type roles: our pa5 as their Ra2, our Nh8 as their Pb5; transferred move h1a1 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0002#3130 (ply 65) [structural distance  $d=40$ , QAP obj=22]: the master played Re8; under  $\mu$  our Rh1 plays the role of that game's Rd1; cross-type roles: our ng8 as their Bb5, our Nh8 as their Pa4; transferred move h1a1 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#3304 (ply 68) [structural distance  $d=38$ , QAP obj=12]: the master played h4; under  $\mu$  our Rh1 plays the role of that game's Ph2 (a **cross-type** correspondence); cross-type roles: our ng8 as their Bd1, our ph4 as their Bd2; transferred move h1a1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ra1 (summed QAP objective 99, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



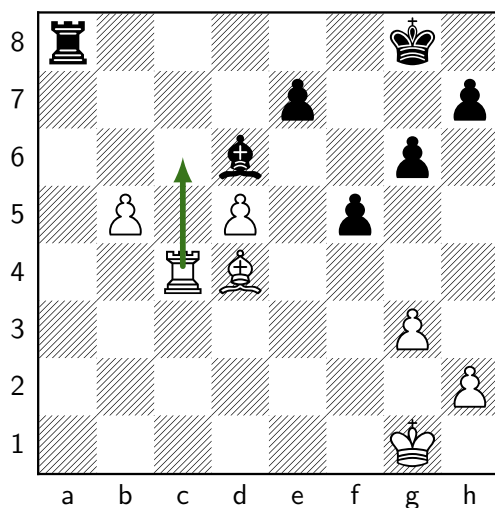
(a)



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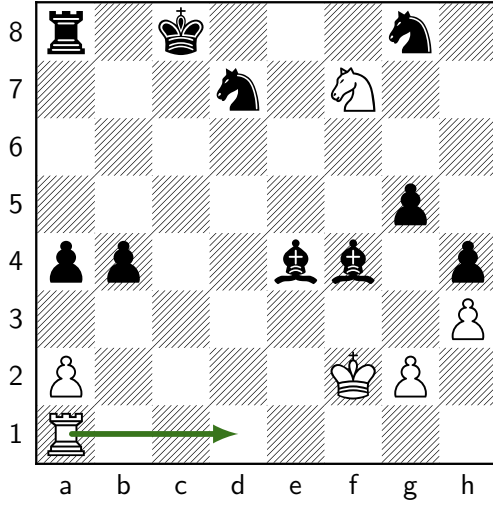


(c)

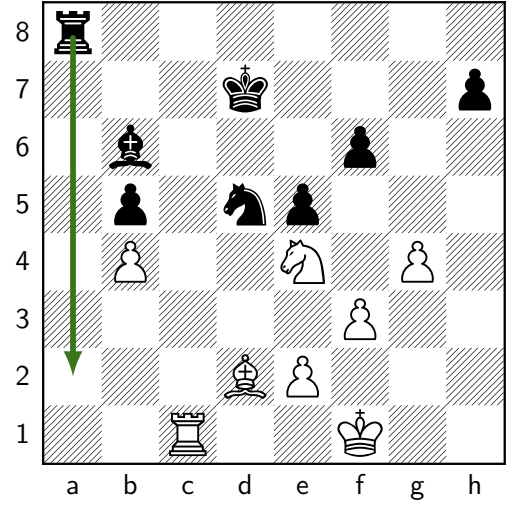


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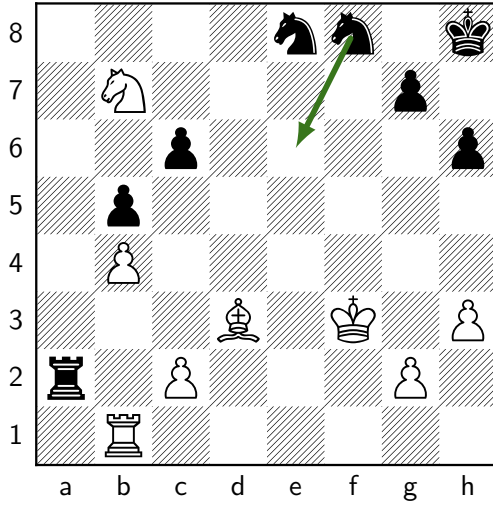
Figure 26: **Move 32, White: Nf7**. (a) Our position; Melanie plays Nf7 (h8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0003#310 (ply 73) [structural distance  $d=31$ , QAP obj=22]: the master played Ne6; under  $\mu$  our Nh8 plays the role of that game’s Nf1; cross-type roles: our pb4 as their Ne8, our Pa2 as their nb7; transferred move h8f7 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#3765 (ply 71) [structural distance  $d=38$ , QAP obj=22]: the master played Rf4; under  $\mu$  our Nh8 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pc5, our ra8 as their pb3; transferred move h8f7 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0003#14 (ply 62) [structural distance  $d=34$ , QAP obj=17]: the master played Rc6; under  $\mu$  our Nh8 plays the role of that game’s Rc4 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Bd4, our Ra1 as their Pd5; transferred move h8f7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Nf7 (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



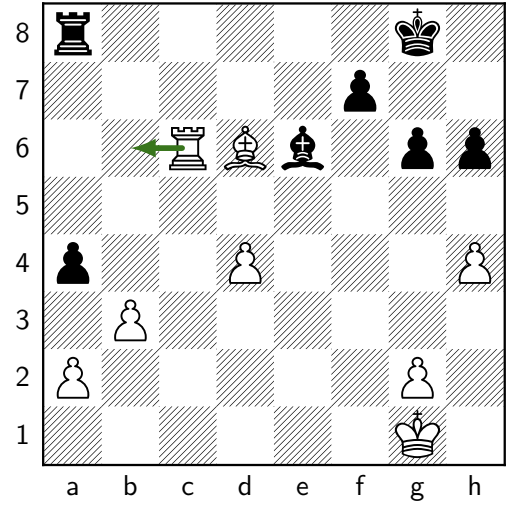
(a)



(b)

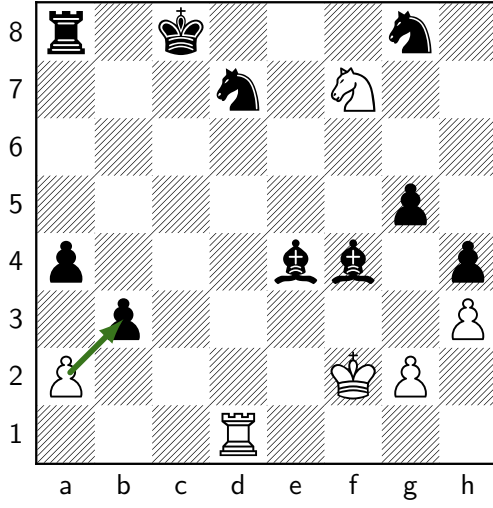


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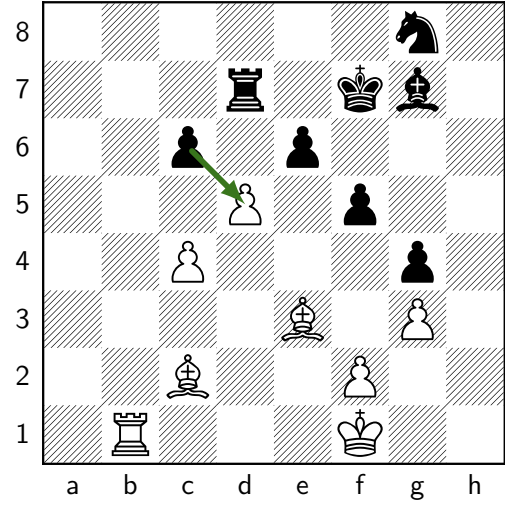


(d)

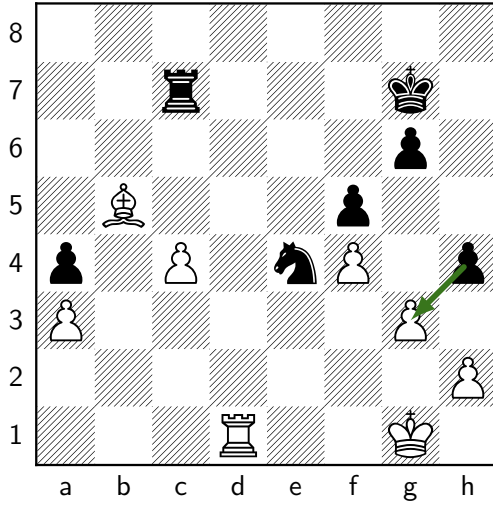
Figure 27: **Move 33, White: Rd1.** (a) Our position; Melanie plays Rd1 (a1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0002#1357` (ply 75) [structural distance  $d=39$ , QAP obj=40]: the master played Ra2; under  $\mu$  our Ra1 plays the role of that game’s Ra1; cross-type roles: our Ph3 as their Bb6, our bf4 as their Pb5; transferred move a1d1 (**endorses this move**). (c) Retrieved from `cctl_top_p0003#310` (ply 73) [structural distance  $d=33$ , QAP obj=40]: the master played Ne6; under  $\mu$  our Ra1 plays the role of that game’s Nf1 (a **cross-type** correspondence); cross-type roles: our ph4 as their Ne8, our Ph3 as their nb7; transferred move a1d1 (**endorses this move**). (d) Retrieved from `cctl_top_p0002#2899` (ply 52) [structural distance  $d=36$ , QAP obj=16]: the master played Rb6; under  $\mu$  our Ra1 plays the role of that game’s Rc6; cross-type roles: our bf4 as their Pa2, our be4 as their Pg2; transferred move a1d1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rd1 (summed QAP objective 108, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



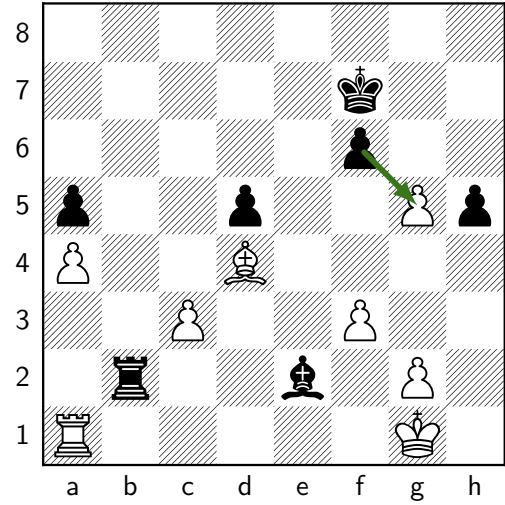
(a)



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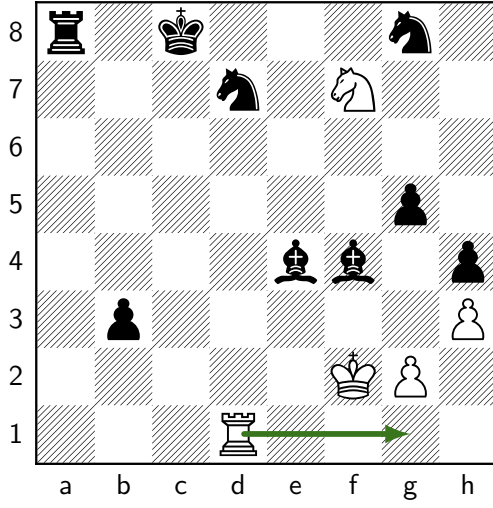


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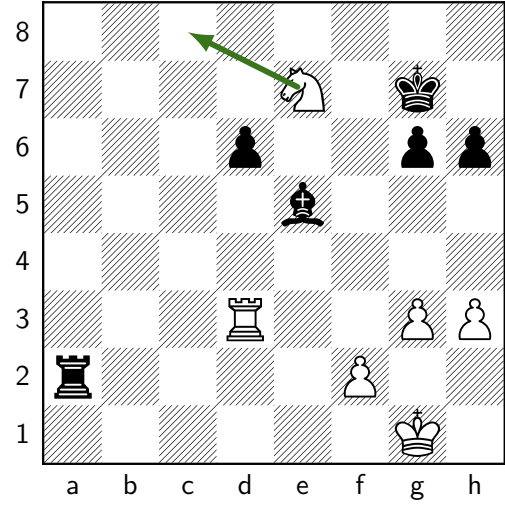


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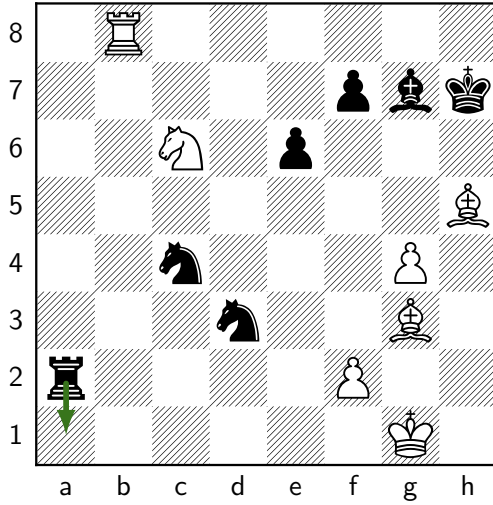
Figure 28: **Move 34, White: axb3.** (a) Our position; Melanie plays axb3 (a2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#703` (ply 81) [structural distance  $d=44$ , QAP obj=88]: the master played `cxd5`; under  $\mu$  our *Pa2* plays the role of that game’s *Pc3*; cross-type roles: our *Rd1* as their *Ng8*, our *Pa2* as their *Rd7*; transferred move `a2b3` (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2758` (ply 77) [structural distance  $d=35$ , QAP obj=64]: the master played `hxc3`; under  $\mu$  our *Pa2* plays the role of that game’s *Ph5*; cross-type roles: our *bf4* as their *Pf5*, our *nd7* as their *pc4*; transferred move `a2b3` (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3424` (ply 77) [structural distance  $d=42$ , QAP obj=54]: the master played `fxg5`; under  $\mu$  our *Pa2* plays the role of that game’s *Pf3*; cross-type roles: our *Rd1* as their *Ph5*, our *pb3* as their *bd4*; transferred move `a2b3` (**endorses this move**). Synthesis: 9 retrieved analogs transferred to `axb3` (summed QAP objective 365, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



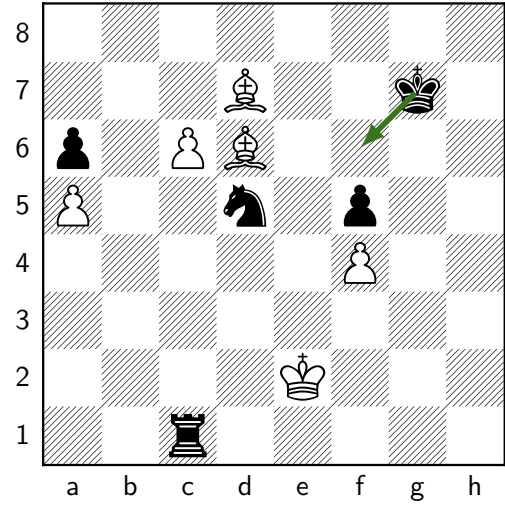
(a)



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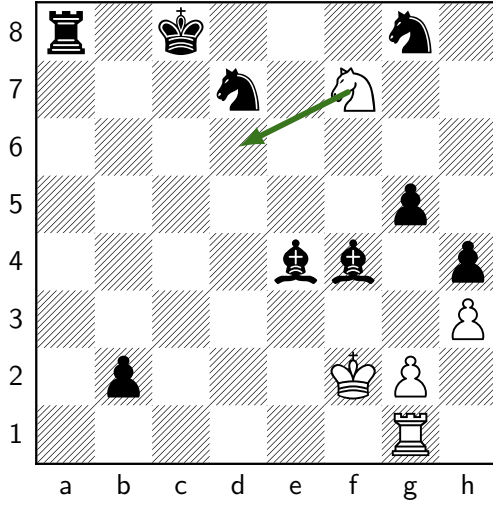


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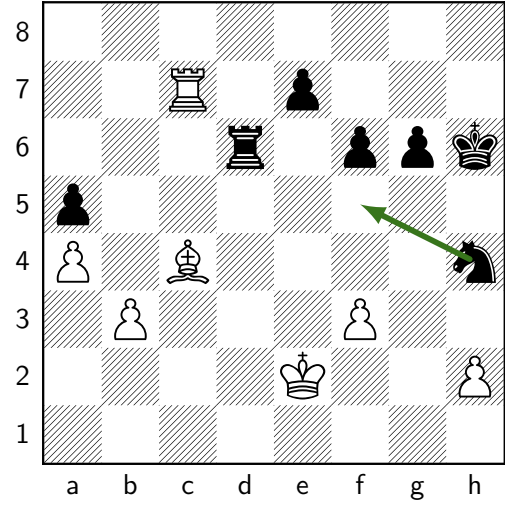


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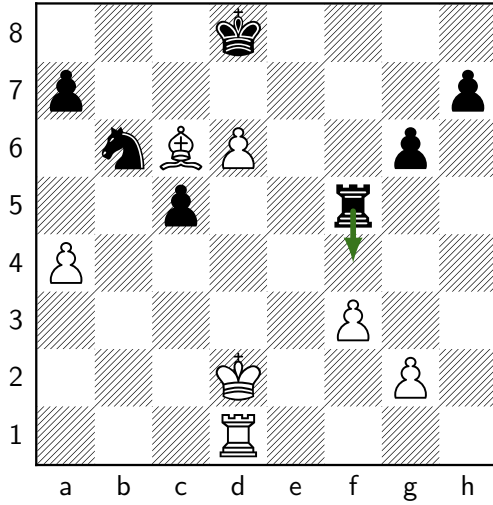
Figure 29: **Move 35, White: Rg1.** (a) Our position; Melanie plays Rg1 (d1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2054` (ply 82) [structural distance  $d=38$ , QAP obj=51]: the master played Nc8; under  $\mu$  our Rd1 plays the role of that game’s Ne7 (a **cross-type** correspondence); cross-type roles: our bf4 as their ra2, our Nf7 as their Rd3; transferred move d1d4. (c) Retrieved from `ccr1_top_p0001#655` (ply 61) [structural distance  $d=33$ , QAP obj=48]: the master played Ra1+; under  $\mu$  our Nf7 plays the role of that game’s Ra7 (a **cross-type** correspondence); cross-type roles: our nd7 as their rb8, our Rd1 as their Bg7; transferred move f7h8. (d) Retrieved from `ccr1_top_p0000#3028` (ply 85) [structural distance  $d=38$ , QAP obj=48]: the master played Kf6; under  $\mu$  our Kf2 plays the role of that game’s Kg2; cross-type roles: our nd7 as their pc6; transferred move f2e1. Synthesis: no retrieved analog transferred to Rg1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



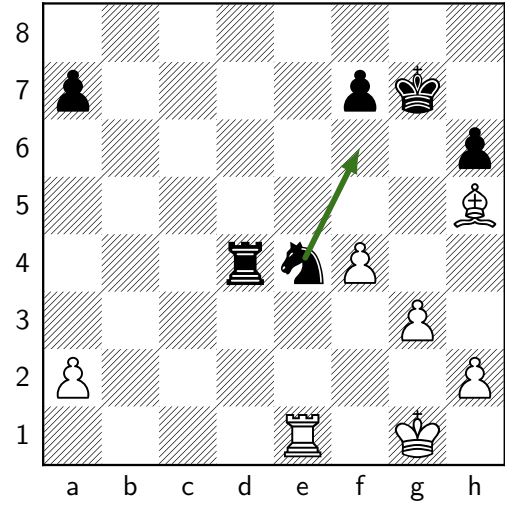
(a)



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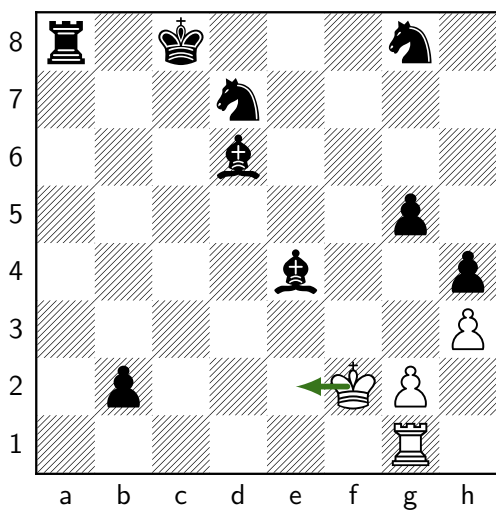


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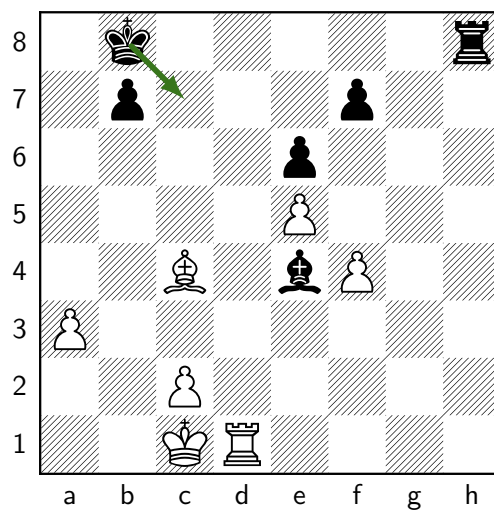


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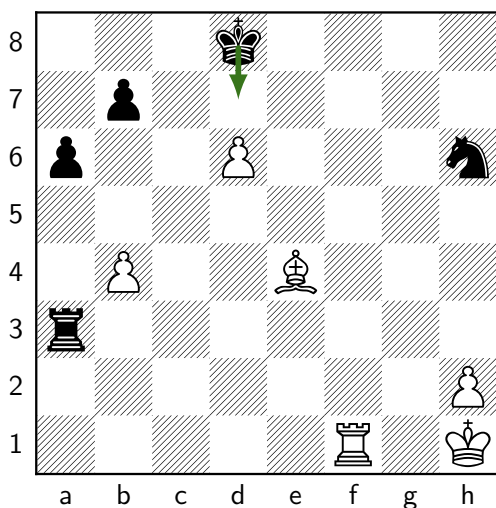
Figure 30: **Move 36, White: Nd6+.** (a) Our position; Melanie plays Nd6+ (f7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2304` (ply 83) [structural distance  $d=40$ , QAP obj=40]: the master played Nf5; under  $\mu$  our Nf7 plays the role of that game’s Nh5; cross-type roles: our be4 as their rc7, our bf4 as their Rd6; transferred move f7d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#356` (ply 67) [structural distance  $d=40$ , QAP obj=39]: the master played Rf4; under  $\mu$  our Nf7 plays the role of that game’s Rf4 (a **cross-type** correspondence); cross-type roles: our be4 as their Nb6, our Rg1 as their bc6; transferred move f7d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1434` (ply 65) [structural distance  $d=41$ , QAP obj=23]: the master played Nf6; under  $\mu$  our Nf7 plays the role of that game’s Ne5; cross-type roles: our Rg1 as their Ph6, our ph4 as their Rd4; transferred move f7d6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nd6+ (summed QAP objective 142, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



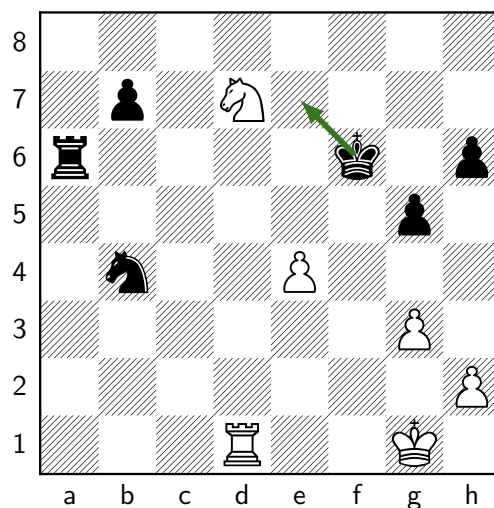
(a)



(b)



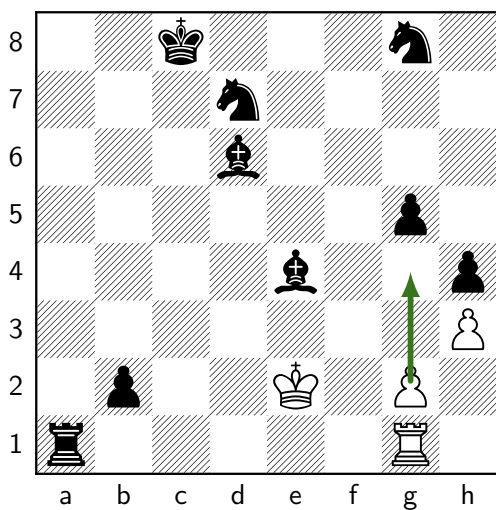
(c)



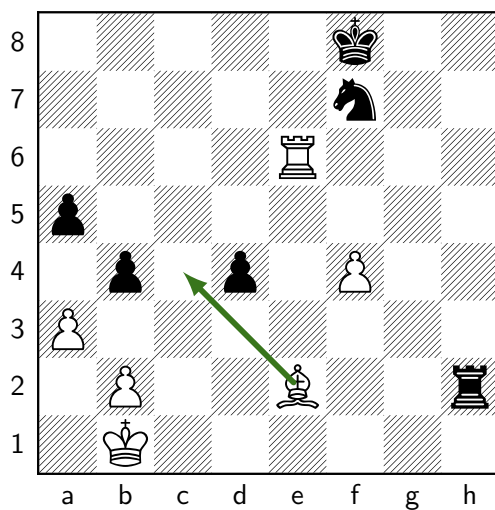
(d)

Figure 31: **Move 37, White: Ke2**. (a) Our position; Melanie plays Ke2 (f2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3249` (ply 65) [structural distance  $d=34$ , QAP obj=29]: the master played Kc7; under  $\mu$  our Kf2 plays the role of that game's Kb1; cross-type roles: our ng8 as their Rh8, our ra8 as their Pb7; transferred move f2e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3788` (ply 79) [structural distance  $d=31$ , QAP obj=16]: the master played Kd7; under  $\mu$  our Kf2 plays the role of that game's Kd1; transferred move f2e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#19` (ply 77) [structural distance  $d=34$ , QAP obj=4]: the master played Ke7; under  $\mu$  our Kf2 plays the role of that game's Kf3; cross-type roles: our be4 as their nd7, our bd6 as their ph2; transferred move f2e2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ke2 (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

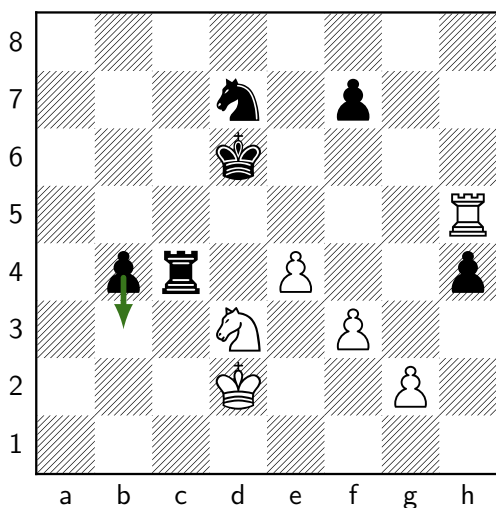




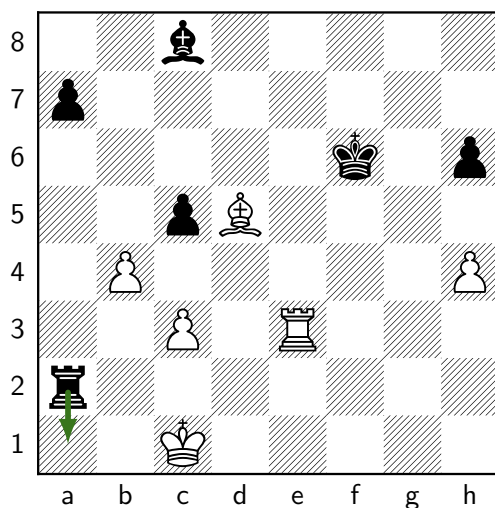
(a)



(b)

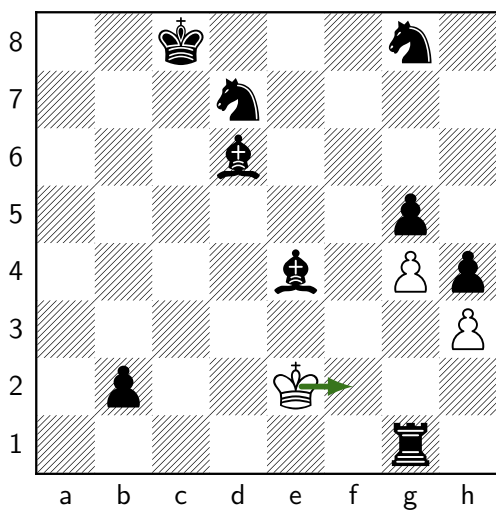


(c)

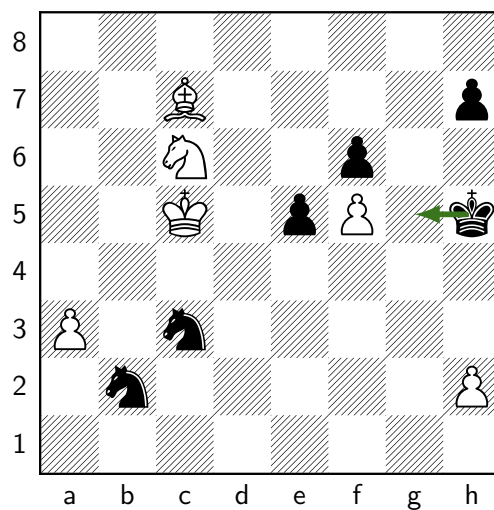


(d)

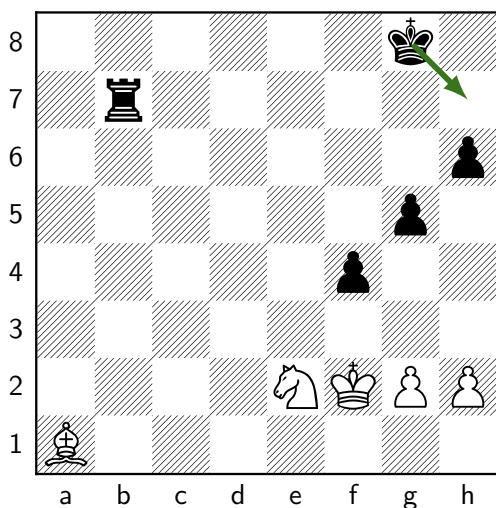
Figure 32: **Move 38, White: g4.** (a) Our position; Melanie plays g4 (g2g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2165` (ply 82) [structural distance  $d=29$ , QAP obj=42]: the master played Bc4; under  $\mu$  our Pg2 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our be4 as their rh2, our Rg1 as their Pa3; transferred move g2g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3361` (ply 83) [structural distance  $d=32$ , QAP obj=27]: the master played b3; under  $\mu$  our Pg2 plays the role of that game’s Pb5; cross-type roles: our pg5 as their Nd7, our ra1 as their pe4; transferred move g2g4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3510` (ply 81) [structural distance  $d=28$ , QAP obj=26]: the master played Ra1+; under  $\mu$  our Pg2 plays the role of that game’s Ra7 (a **cross-type** correspondence); cross-type roles: our Rg1 as their Pc5, our ra1 as their pb4; transferred move g2g4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to g4 (summed QAP objective 138, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



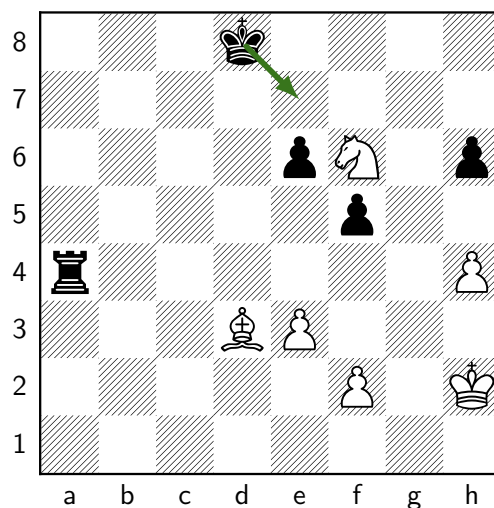
(a)



(b)

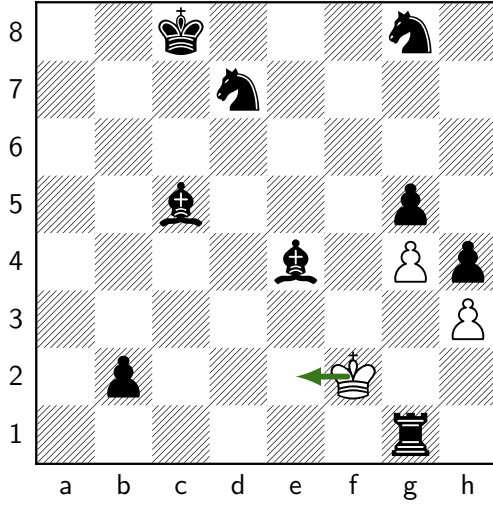


(c)

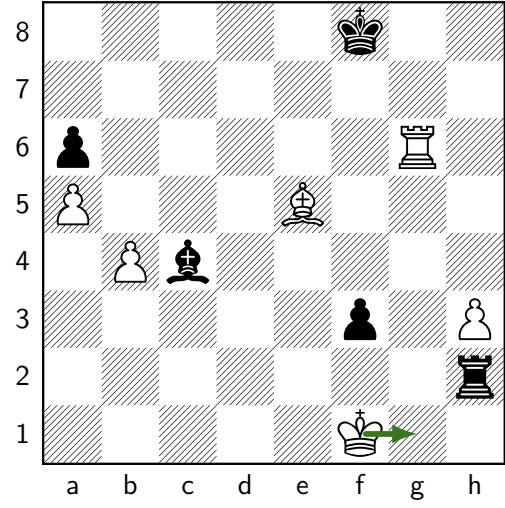


(d)

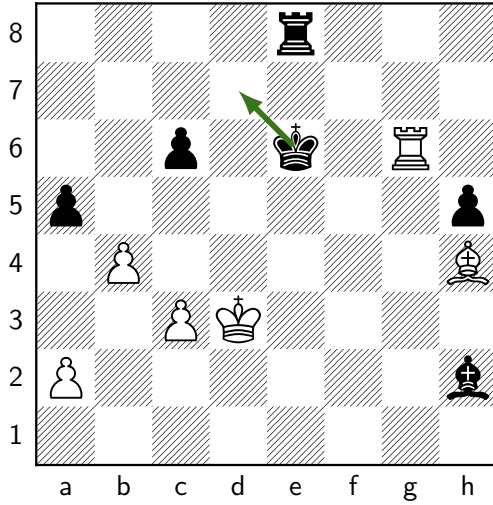
Figure 33: **Move 39, White: Kf2**. (a) Our position; Melanie plays Kf2 (e2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3790` (ply 89) [structural distance  $d=22$ , QAP obj=23]: the master played Kg5; under  $\mu$  our Ke2 plays the role of that game’s Kh4; cross-type roles: our rg1 as their bc7, our rg1 as their nc6; transferred move e2f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3751` (ply 79) [structural distance  $d=23$ , QAP obj=15]: the master played Kh7; under  $\mu$  our Ke2 plays the role of that game’s Kg1; cross-type roles: our be4 as their Rb7, our rg1 as their ne2; transferred move e2f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#751` (ply 79) [structural distance  $d=24$ , QAP obj=14]: the master played Ke7; under  $\mu$  our Ke2 plays the role of that game’s Kd1; cross-type roles: our pg5 as their nf6, our nd7 as their Ph6; transferred move e2f2 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Kf2 (summed QAP objective 123, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



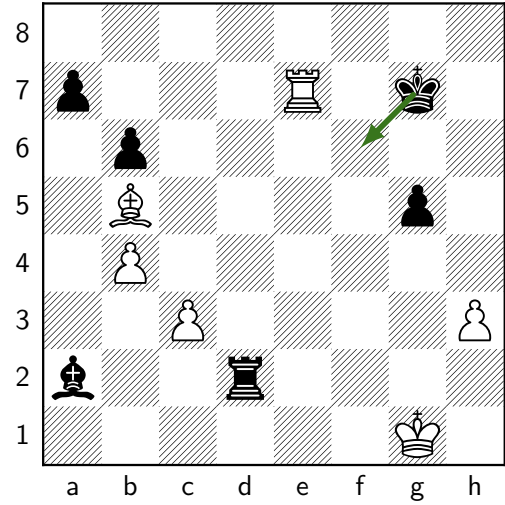
(a)



(b)

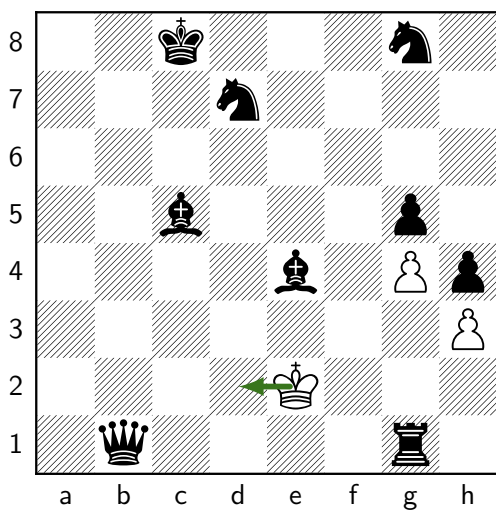


(c)

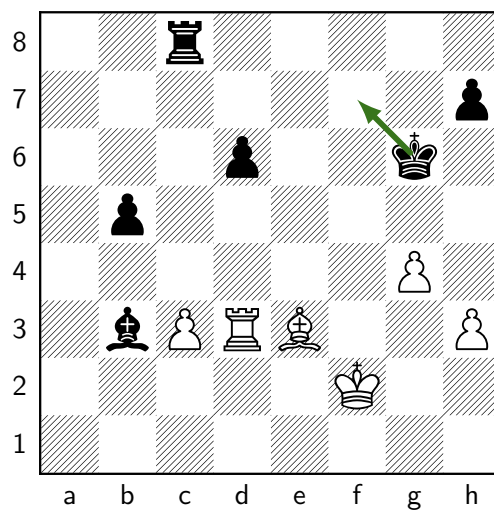


(d)

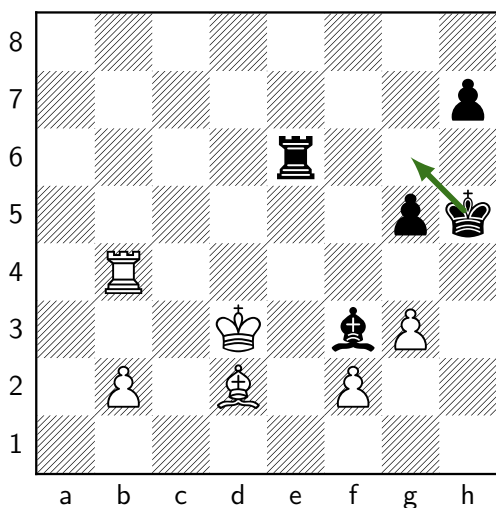
Figure 34: **Move 40, White: Ke2**. (a) Our position; Melanie plays Ke2 (f2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1335` (ply 84) [structural distance  $d=30$ , QAP obj=69]: the master played Kg1; under  $\mu$  our Kf2 plays the role of that game’s Kf1; cross-type roles: our nd7 as their Rg6; transferred move f2e2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1945` (ply 73) [structural distance  $d=31$ , QAP obj=47]: the master played Kd7; under  $\mu$  our Kf2 plays the role of that game’s Ke3; cross-type roles: our be4 as their Re8, our bc5 as their rg6; transferred move f2e2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1682` (ply 91) [structural distance  $d=32$ , QAP obj=46]: the master played Kf6; under  $\mu$  our Kf2 plays the role of that game’s Kg2; cross-type roles: our bc5 as their re7, our pg5 as their bb5; transferred move f2e2 (**endorses this move**). Synthesis: 25 retrieved analogs transferred to Ke2 (summed QAP objective 464, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



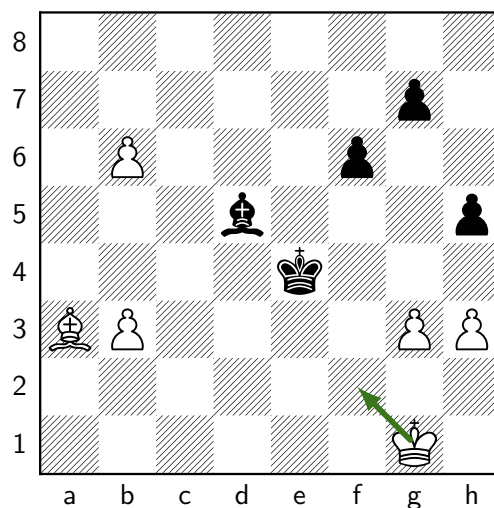
(a)



(b)



(c)



(d)

Figure 35: **Move 41, White: Kd2.** (a) Our position; Melanie plays Kd2 (e2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3090` (ply 75) [structural distance  $d=60$ , QAP obj=21]: the master played Kf7; under  $\mu$  our *Ke2* plays the role of that game’s *Kg3*; cross-type roles: our *be4* as their *Pb5*, our *bc5* as their *pg4*; transferred move e2d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2991` (ply 89) [structural distance  $d=58$ , QAP obj=21]: the master played Kg6; under  $\mu$  our *Ke2* plays the role of that game’s *Kh4*; cross-type roles: our *nd7* as their *Ph7*, our *ph4* as their *Re6*; transferred move e2d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2119` (ply 92) [structural distance  $d=60$ , QAP obj=14]: the master played Kf2; under  $\mu$  our *Ke2* plays the role of that game’s *Kg1*; cross-type roles: our *qb1* as their *Ba3*, our *rg1* as their *bd5*; transferred move e2d2 (**endorses this move**). Synthesis: 19 retrieved analogs transferred to Kd2 (summed QAP objective 157, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.